Course Name: Computer Organization

Course Outcome

CO1- Understand the basics of digital computer system.
CO2- Demonstrate the principle of arithmetic operations on unsigned, signed integers and floating point numbers.
CO3- Understand the concepts of combinational and sequential circuits and their applications.
CO4- Understand the CPU architecture and organization.
CO5- Explain the basic concepts of pipelining.
CO6- Design the steps for the execution of the complete instruction for hardwired and micro-programmed control unit.
CO7- Explain the function of memory hierarchy.
CO8- Determine the interface of CPU with input/output devices and their modes of transfer.

Printed Pages: 03 **University Roll No. .....**

End Term Examination, Odd Semester 2023-24
B. Tech. (CS/AIML/CCV/CSF/DA/IIOT), II Year, III Semester
BCSC 1005 : Computer Organization

Time: 3 Hours **Maximum Marks: 50**

Section – A

*Attempt **All** Questions* 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	Bl.	KL
1	Consider two binary numbers $A_3A_2A_1A_0$ and $B_3B_2B_1B_0$ are stored in two registers A & B based on Parallel-In Parallel-Out mode respectively. Design a combinational logic to subtract two numbers.	4	3	U	C
2	Show how a computer processor evaluates the following arithmetic expression by using reverse polish notation? ((54-20)*((100-3*10)/7)/2)/2+15	4	4	A	P
3	Represent $(175.51)_2$ in single and double precision floating point number.	4	2	A	P
4	Consider a cache consisting of 256 blocks of 16 words each for a total of 4096 words. Also, assume that the main memory is addressable by a 16 bits address and accommodate 12 bits data. How many bits are there in each of Tag, Index, Block, and Word field for 2-way set associative technique? Include a valid bit. What is the size of the cache memory?	4	6	U	P
5	Consider a pipeline having 4 phases with duration 60, 50, 90 and 80 ns. Given latch delay is 10 ns. Calculate-	4	5	U	P

	1. Pipeline cycle time 2. Non-pipeline execution time 3. Speed up ratio 4. Pipeline time for 1000 tasks 5. Sequential time for 1000 tasks 6. Throughput Or In a certain scientific computation, it is necessary to perform the following arithmetic expression. $A_i * \left(\frac{B_i}{C_i}\right) + D_i$ (a) Specify the pipeline configuration to carry out the above task. (b) List the content of all registers in a space-time diagram for 5 tasks.				

Section – B

*Attempt **All** Questions* 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Define (a) Locality of reference (b) Write-back (c) Page fault	3	5	R	F
7	A computer uses RAM chips of 1024x1capacity. (a) How many chips are needed, and how should their address lines be connected to provide a memory capacity of 1024 bytes? (b) How many chips are needed to provide a memory capacity of 16K bytes?	3	7	U	C
8	Define (a) Virtual memory (b) Instruction cycle (c) Interrupt	3	6	R	F
9	Explain different methods of asynchronous data transfer.	3	8	R	F
10	What sequence of micro-operations is performed by a computer system to execute the STA instruction for which two references to the memory are required to compute the memory address?	3	7	U	C

Section – C

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	A virtual memory system has an address space of 8K words, a memory space of 4K words, and page & block sizes are of 1K words. The following page reference changes occur during a given time interval. 3 0 2 1 2 1 3 4 0 1 0 3 2 5 6 0. Determine the four pages that are resident in main memory after each page reference change if the replacement algorithm used is (a) FIFO (b) LRU	5	7	U	P
12	Explain a DMA controller and its mode of data transfer with proper illustration.	5	8	R	F
13	A computer employs RAM chips of 256x8 and ROM chips of 1024x8. The computer system needs 2K bytes of RAM, 4K bytes of ROM, and four interface units each with four registers. A memory-mapped I/O configuration used. The two highest-order bits of the address bus are assigned 00 for RAM, 01 for ROM, and 10 for interface registers. (a) How many RAM and ROM chips are needed? (b) Draw a memory-address map for the system.	5	7	U	P

SOFTWARE ENGINEERING

CO1: Understand the basic concepts of software engineering.
CO2: Apply software processes to solve real world problems.
CO3: Estimate the cost, effort and schedule of software using COCOMO Model.
CO4: Analyze the software design techniques (structure chart, SDM, sequence diagram).
CO5: Understand the basic concepts of OO analysis and design.
CO6: Develop the test cases to validate the software.
CO7: Understand the basic models of software Quality and maintenance

Printed Pages: 4 **University Roll No. .....**

End Term Examination, Odd Semester 2023-24
B.Tech., 2 Year, III Semester
BCSC 0009 & Software Engineering

Time: 3 Hours **Maximum Marks: 50**

Instruction for students:

1. Attempt all questions
2. Do not switch between section A and Section B; first, solve one section, then move to second
3. Use only a blue pen to answer
4. No B sheet will be given

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	For the scenario described below, which life cycle model would you choose? Give the reason why you would choose this model. You are interacting with the MIS department of a very large oil company with multiple departments. They have a complex regency system. Migrating the data from this legacy system is not an easy task and would take a considerable time. The oil company is very particular about processes, acceptance criteria and legal contracts	4	2	An	C
2	Suppose that a project was estimated to be 400 KLOC. Calculate the effort and development time for two modes i.e., organic and semidetached.	4	1	R	C
3	An application has the following:10 low external inputs, 12 high external outputs, 20 low internal	4	3	A	P

	logical files, 15 high external interface files, 12 average external inquiries and a value adjustment factor of 1.10. What is the unadjusted and adjusted function point count?				
4	What is the purpose of data flow diagrams? What are the notations used for the same? Explain by constructing a context flow diagram level -0 DFD and level-1 DFD for a library management system?	4	5	A	M
5	Consider the process of ordering a pizza over the phone. Draw the use case diagram and also sketch the activity diagram representing each step of the process, from the moment you pick up the phone to the point where you start eating the pizza. Include activities that others need to perform. Add exception handling to the activity diagram you developed. Consider at least two exceptions. (Ex: Delivery person wrote down wrong address, deliver person brings wrong pizza). OR Consider the process of ordering a pizza over the phone. Draw the class diagram and also sketch the activity diagram representing each step of the process, from the moment you pick up the phone to the point where you start eating the pizza. Include activities that others need to perform. Add exception handling to the activity diagram you developed. Consider at least two exceptions. (Ex: Delivery person wrote down wrong address, deliver person brings wrong pizza).	4	5	A	M

Section – B

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Annual Change Traffic (ACT) for a software system is 15% per year. The development effort is 600 PMs. Compute estimate for Annual Maintenance Effort	3	7	A	P

	(AME). If life time of the project is 10 years, what is the total effort of the project?				
7	While conducting software testing, what testing principles does the software engineer need to follow?	3	6	R	C
8	You have an approaching deadline but lack data to complete a project. How would you handle this?	3	6	U	P
9	Define: Regression testing. Distinguish: top-down and bottom-up integration. How is testing different from debugging?	3	5	E	C

10	Develop a Control Flow Graph of the code given below and find the cyclomatic complexity?
	IF A = 10 THEN
	IF B > C THEN
	A = B
	ELSE
	A = C
	ENDIF
	ENDIF
	Print A
	Print B
	Print C
		3	6	A	P

Section – C

*Attempt **All** Questions*

5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	You started a software house two years ago that builds and implements custom software solutions for clients. As there existed no organizational project management standard and unified processes, your company relied on senior project managers capable of managing projects and delivering software to clients based on their own approaches and experience. Which of the following is the maturity level that best describes your company in terms of CMMI? Justify your answer and also discuss CMMI model in detail.	5	7	A	C
12	In software maintenance tackling the changes in the hardware or software environment where the	5	7	R	C

	software works, is comes under which types of maintenance? Justify your answers.				
13	Consider a program for determining the Previous date. Its input is a triple of day, month and year with the values in the range $1 \leq \text{month} \leq 12$ $1 \leq \text{day} \leq 31$ $1900 \leq \text{year} \leq 2025$ The possible outputs would be Previous date or invalid input date. Design the boundary value test cases.	5	6	An	C

Course Name: Database Management System

Course Outcome

- CO1: Understand the concept of database management systems and Relational database.
- CO2: Identify the various data model used in database design.
- CO3: Design conceptual models of a database using ER modeling for real life applications and construct queries in Relational Algebra.
- CO4: Create and populate a RDBMS for a real life application, with constraints and keys using SQL.
- CO5: Select the information from a database by formulating complex queries in SQL.
- CO6: Analyze the existing design of a database schema and apply concepts of normalization to design an optimal database.
- CO7: Discuss indexing mechanisms for efficient retrieval of information from a database.
- CO8: Discuss recovery system and be familiar with introduction to web database, distributed databases.

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End Term Examination, Odd Semester, 2023-24
B. Tech. (CSE), Year: II, Semester: III
BCSC1003: DATABASE MANAGEMENT SYSTEM

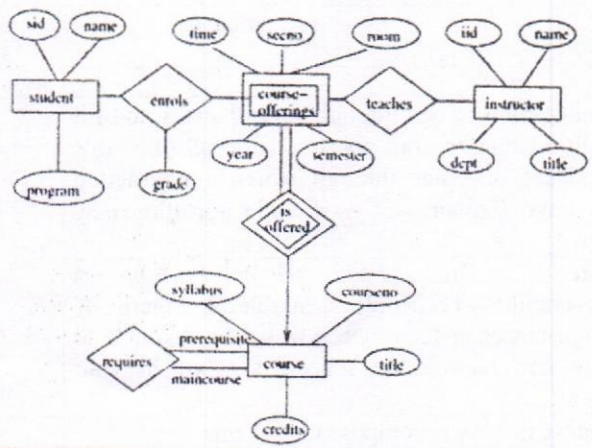
Time: 3 Hours **Maximum Marks: 50**

Instruction for students:

1. Attempt all questions
2. Don't switch between the section A, B and Section C, first solve one section then move to second
3. Use only blue pen to answer
4. No B sheet will be given

Section – A

*Attempt **All** Questions* 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Explain the 3-Level architecture of Database Management Systems using proper diagram.	4	CO1	U	C
2	Construct appropriate tables for the given E-R diagram. 	4	CO2	C	P

3	Given a relational Schema R (A, B, C, D) and set of Function Dependency FD = {B→A, AD→BC, C→ABD}. Find the canonical cover?	4	CO3	E	P
4	Consider the following two transactions: T1: read (*A*); read (*B*); **if** A = 0**then** *B*: = *B* + 1; write(*B*). T2: read (*B*); read (*A*); **if** B = 0**then** *A*: = *A* + 1; write (*A*). Add lock and unlock instructions to transactions T1 and T2, so that they observe the two-phase locking protocol.	4	CO5	A	P
5	Sometimes a transaction has to be undone after it has committed, because it was erroneously executed, for example because of erroneous input by a bank teller. Give an example to show that using the normal transaction undo mechanism to undo such a transaction could lead to an inconsistent state.	4	CO5	An	F

Section – B

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	A transaction passes through several states, until it finally commits or aborts. List all possible sequences of states through which a transaction may pass. Explain why each state transition may occur.	3	CO5	U	F
7	What is a recoverable schedule? Why is recoverability of schedules desirable? Are there any circumstances under which it would be desirable to allow non recoverable schedules? Explain your answer.	3	CO8	An	C
8	Suppose that we decompose the schema R = (A, B, C, D, E) into (A, B, C) and (A, D, E).	3	CO6	E	M

	Show that this decomposition is a lossless-join decomposition if the following set F of functional dependencies holds: {A → BC, CD → E, B → D, E → A}				
9	Define Boyce-Codd normal form. How does it differ from 3NF? Why is it considered a stronger form of 3NF?	3	CO6	U	C
10	Check whether the given schedule is conflict serializable or not? If not, is it view equivalent? **S** **T1** | **T2** | **T3** R (A) W (A) W (A) W (A)	3	CO7	An	P

Section – C

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	Consider the concurrent execution of two transactions, T1 & T2 as: **T1** | **T2** R (x) x = x – N R (x) x = x + M W (x) R (y) W (x) y = y + N W (y)	5	CO7	E	P

	The initial values of x, y, M & N are 200, 100, 10 & 20 respectively. Now determine: (a). What would be the values of x & y at the end of the execution of T1 & T2? (b). If the transactions are executed in serial manner; what would be the values of x & y at the end of the serial execution of T1 & T2?				
12	What is a fragment of a relation? What are the main types of fragments? Why is fragmentation a useful concept in distributed database design?	5	CO8	U	C
13	Consider the following relations for an order-processing application database at ABC, Inc. ORDER (O#, Odate, Cust#, Total_amount) ORDER_ITEM(O#, I#, Qty_ordered, Total_price, Discount%) Assume that each item has a different discount. The Total_price refers to one item, Odate is the date on which the order was placed, and the Total_amount is the amount of the order. If we apply a natural join on the relations ORDER_ITEM and ORDER in this database, what does the resulting relation schema look like? What will be its key? Show the FDs in this resulting relation. Is it in 2NF? Is it in 3NF? Why or why not? (State assumptions, if you make any.)	5	CO6	C	P

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End-Term Examination, Odd Semester 2023-24

B.Tech. CSE, II : Year, III : Semester

BCSC 0081:Blockchain Essentials

Time: 1.5-hour **Maximum Marks: 50**

SECTION A

Note: Attempt All Questions. **30 x 1 =30 Marks**

1. How do you access a specific character in a string in JavaScript?

A) By using the length property and the index of the character

B) By using the indexOf() method and the index of the character

C) By using square brackets and the index of the character

D) By using the charAt() method and the index of the character

2. How do you find the length of a string in JavaScript?

A) By using the length() method

B) By using the size() method

C) By using the count() method

D) By using the length property

3. How do you concatenate two strings in JavaScript?

A) By using the + operator

B) By using the concat() method

C) By using the join() method

D) By using the append() method

4. Spaces and line breaks are not important.
A) TRUE
B) FALSE
C) Can be true or false
D) Can not say

5. What is true about Arrays?
A) JavaScript does not support associative arrays
B) You should use objects when you want the element names to be strings
C) You should use arrays when you want the element names to be numbers
D) All of the above

6. The _____ method also joins all array elements into a string.
A) toString()
B) join()
C) pop()
D) shift()

7. Arrays always use numbered indexes.
A) Yes
B) No
C) Can be yes or no
D) Can not say

8. The capability to store one object inside another object known as?
A) Encapsulation
B) Aggregation
C) Inheritance

D) Polymorphism

9. The syntax for adding a property to an object is?
A) objectProperty == propertyValue;
B) objectProperty = propertyValue;
C) objectName.objectProperty = propertyValue;
D) objectName.objectProperty == propertyValue;

10. The _____ operator is used to create an instance of an object.
A) this
B) self
C) find
D) new

11. In REST, what does the HTTP POST method typically represent?
A) Retrieving data
B) Updating existing data
C) Creating new data
D) Deleting data

12. What is the purpose of the HTTP status code 404 in a RESTful API response?
A) OK - Request Successful
B) Bad Request
C) Not Found
D) Unauthorized

13. What does CORS (Cross-Origin Resource Sharing) address in the context of RESTful APIs?
A) Cache control

B) Authentication
C) Cross-origin request security
D) Rate limiting

14. What is the purpose of URL parameters in a RESTful API endpoint?
A) To include additional headers
B) To specify the request method
C) To pass data in the request body
D) To provide input values for the resource

15. What is the main factor that determines the value of fiat currency?
A) The supply and demand for the currency
B) The amount of gold backing the currency
C) The interest rates set by the government or central bank
D) The inflation rate

16. What is the role of central banks in managing paper money?
A) To set the value of the currency
B) To prevent counterfeiting
C) To adjust interest rates and the money supply
D) To determine the supply and demand for the currency

17. What is one of the main advantages of using gold as money?
A) It is subject to inflation
B) It is a digital asset
C) It is easy to store and transport
D) It retains its value over time

18. What is the role of money in a modern economy?
A) To facilitate transactions and exchanges
B) To create wealth
C) To provide a means of production

D) None of the above

19. What is the advantage of using digital currency?
A) It is more secure than other forms of currency
B) It is easier to use than other forms of currency
C) It is more widely accepted than other forms of currency
D) All of the above

20. Diversification of one's stock portfolio is important because
A) it reduces the likelihood of a dramatic change in value over a relatively short period of time.
B|) it will always yield a higher rate of return than a narrow group of related stock investments.
C) the dividends of a diversified stock portfolio are guaranteed.
D) the initial funds used to purchase the stocks will always be returned to the Investor.

21. Which of the following will help you generate income and accumulate wealth?
A) Getting another credit card so you will have more borrowing power in the future.
B) Borrowing funds in order to increase your current consumption .
C) Allocating funds regularly into savings and investments.
D) Making only the minimum monthly payment on your outstanding credit card balances.

22. If you are using a credit card prudently, you should
A) make only the minimum payment each month.
B) pay the balance in full each month.
C) pay only the interest on any outstanding balance.
D) gradually expand the amount borrowed to the card's credit limit.

23. Personal financial success is primarily the result of
A) generous welfare and unemployment programs.

B) spending more than you earn.
C) doing what you are passionate about regardless of whether others value it.
D) providing services others value while working hard to achieve financial Goals.

24. Should you accept your employer's offer to match eligible earnings used to purchase company stock?
A) Yes, you should purchase the company stock and hold onto it for as long as possible.
B) Probably, the matching of your contributions makes it attractive, but if you buy the company stock you should sell it when permitted to do so in order to keep your portfolio diversified.
C) No, you should not participate. The company is merely trying to take advantage of you.
D) No. It never makes sense to buy the stock of the company for which you Work.

25. What is the purpose of the "Bearer" in an Authorization header for token-based authentication in a RESTful API?
A) It identifies the user making the request
B) It specifies the encryption algorithm used
C) It indicates the type of token being used
D) It denotes the access level of the token

26. Which of the following is a common format for paginating results in RESTful API response?
A) JSON-LD
B) HAL (Hypertext Application Language)
C) GraphQL
D) Link headers

27. Which of the following scenarios is an example of a "length extension attack"?
A) Finding a collision for a given hash value

B) Extending the length of the hash output
C) Calculating the hash of a message without knowing the original input
D) Forging a valid message with an extended hash

28. In a Merkle-Damgård construction, what is the purpose of the finalization step?
A) Padding the message
B) Generating the initial hash value
C) Applying the compression function
D) Reversing the hash output

29. What is the purpose of using salt in password hashing?
A) To make the hash function faster
B) To increase the chances of collisions
C) To add a secret key for encryption
D) To protect against rainbow table attacks

30. Which mathematical structure forms the basis of ECC encryption?
A) Prime numbers
B) Elliptic curves
C) RSA modulus
D) Fibonacci sequences

SECTION B

Note: Attempt All Questions. **40 x 0.5 =20 Marks**

31. _____ serializes only the enumerable own properties of an object.
A) JSON.Parse()
B) JSON.Stringify()
C) JSON.Null()
D) JSON.Objectify()

32. Object ..... is the process of converting an objects state to a string from which it can later be restored.
A) prototype
B) class
C) serialization
D) extensible

33. What is the purpose of the "block timestamp" in a Bitcoin block?
A) To indicate the time the block was mined
B) To specify the Bitcoin price at the time of mining
C) To identify the miner's location
D) To set the difficulty level for mining

34. What is the term for the reward given to miners for adding a new block to the Bitcoin blockchain?
A) Transaction fee
B) Block subsidy
C) Coinbase reward
D) Confirmation fee

35. What is the main purpose of the Bitcoin blockchain?

A) To store cryptocurrency wallets

B) To record transactions and their sequence

C) To mine new bitcoins

D) To perform decentralized computations

36.The web development environment (JavaScript) offers which standard construct for data validation of the input entered by the user.
A) Controlled loop constructs
B) Server page access
C) Client side Event

D) Permit server-side

37. JavaScript is ideal to _____
A) make computations in HTML simpler
B) minimize storage requirements on the web server
C) increase the download time for the client
D) increase the loading time of the website

38. What is the recommended key length for RSA encryption to provide a reasonable level of security as of the time of my knowledge cutoff in September 2021?
A) 64 bits
B) 128 bits
C) 256 bits
D) 2048 bits

39. A hexadecimal literal begins with _____

A) 00

B) 0x

C) 0X

D) Both 0x and 0X

40. Which of the following is not considered as an error in JavaScript?
A) Syntax error
B) Missing of semicolons
C) Division by zero
D) Missing of Bracket

41. Is Bootstrap3 mobile-first?
A) True

B) False
C) Can't say
D) May be

42. In ECC, the security of the encryption relies on the difficulty of solving which mathematical problem?
A) Integer factorization
B) Discrete logarithm
C) Quadratic residue
D) Modular addition

43. The statement a===b refers to _____
A) Both a and b are equal in value, type and reference address
B) Both a and b are equal in value
C) Both a and b are equal in value and type
D) There is no such statement.

44. What is the purpose of a timestamp in a digital signature scheme?
A) To prove the age of the digital signature
B) To increase the security of the private key
C) To encrypt the digital signature
D) To add a watermark to the signed document

45. In a digital signature scheme, what role does the private key play in the signing process?

A) It verifies the signature.

B) It generates the hash of the message.

C) It encrypts the message.

D) It creates the digital signature.

46. What is the primary purpose of a digital signature in cryptography?

A) Data encryption

B) Data compression

C) Data integrity verification

D) Data obfuscation

47. In ECC, what is the primary disadvantage compared to traditional RSA encryption?

A) Slower encryption and decryption

B) Larger key sizes for equivalent security

C) Vulnerability to quantum attacks

D) Limited compatibility with legacy systems

48. In ECC, what is the significance of "cofactor" in elliptic curve group operations?

A) It defines the number of points on the curve.

B) It specifies the size of the public key.

C) It affects the efficiency of point multiplication.

D) It determines the prime number used in ECC.

49. In ECC, what role does the "twist security" property play in ensuring the security of the cryptosystem?

A) It ensures that ECC keys are rotated periodically for better security.

B) It prevents attackers from "twisting" the elliptic curve to break encryption.

C) It guarantees that ECC keys are kept secret.

D) It has no direct relevance to ECC security.

50. ECC is known for its efficiency in terms of key size compared to traditional RSA encryption. Approximately, how much smaller is an ECC key for equivalent security compared to an RSA key?

A) ECC keys are about the same size as RSA keys.

B) ECC keys are half the size of RSA keys.

C) ECC keys are one-quarter the size of RSA keys.

D) ECC keys are one-eighth the size of RSA keys.

51. In ECC, what is a "point at infinity," and how is it related to the mathematical structure of elliptic curves?

A) It's a point with an undefined value; it has no relevance in ECC.

B) It's a point representing an infinitely large value; it's used in ECC for faster computations.

C) It's a point that serves as the identity element in ECC group operations.

D) It's a special point that reveals the private key in ECC if used improperly.

52. What mathematical property of elliptic curves makes ECC resistant to Shor's algorithm, a quantum algorithm that threatens traditional public-key encryption methods?

A) Point doubling

B) The elliptic curve discrete logarithm problem (ECDLP)

C) Prime number generation

D) Modular arithmetic

53. Which of the following is not an argument of arrange method?
A) start
B) stop
C) end
D) add

54. Which of the method is used to check the dimension of an array
A) Shape
B) dim
C) shape
D) Dim

55. What is the recommended key length for ECC encryption to provide a reasonable level of security?

A) 64 bits

B) 128 bits

C) 256 bits

D) It varies depending on the curve used.

56. Which ECC-based encryption algorithm is commonly used for secure communications over the internet, including HTTPS connections?

A) ECDSA (Elliptic Curve Digital Signature Algorithm)

B) ECDH (Elliptic Curve Diffie-Hellman)

C) ECIES (Elliptic Curve Integrated Encryption Scheme)

D) ECCC (Elliptic Curve Cryptographic Cipher)

57. Which key is kept secret in ECC encryption?

A) Public key

B) Private key

C) Both public and private keys

D) Neither public nor private keys

58. What is the primary advantage of ECC encryption over traditional RSA encryption?

A) Faster encryption and decryption

B) Smaller key sizes for equivalent security

C) Simpler key management

D) Resistance to quantum attacks

59. Which module in Node.js is commonly used for handling HTTP requests and responses?
A) fs
B) http
C) path

D) url

60. Which of the following class in Bootstrap is used to create a large button?
A) .btn-xl
B) .btn-lrg
C) .btn-large
D) .btn-lg

61. The class in Bootstrap which is used to specify the collapsible elements is -
A) .collapse
B) .carousel
C) .pager
D) None of the above

62. A class is a type of function, but instead of using the keyword function to initiate it, which keyword do we use?
A) Constructor
B) Class
C) Object
D) DataObject

63. What are the two ways to handle data in React?
A) State & Props
B) Services & Components
C) State & Services
D) State & Component

64. Which of the following class in Bootstrap is used to create a badge?
A) .tag
B) .badge
C) .page
D) .flag

65. How many numbers of elements a valid react component can return?
A) 1
B) 2
C) 4
D) 5

66. What does CRUD stand for in the context of web development?
A) Create, Replace, Update, Delete
B) Create, Retrieve, Update, Delete
C) Copy, Read, Update, Drop
D) Construct, Rearrange, Utilize, Drop

67. Which React hook is commonly used for managing state in a functional component?
A) useState
B) useEffect
C) useContext
D) useReducer

68. In React, what is the role of the map function when rendering a list of items?
A) Filtering the list
B) Sorting the list
C) Transforming each item in the list
D) Removing duplicate items

69. Which of the following is the correct way to import the fs module in Node.js?
A) import fs from 'fs';
B) const fs = require('fs');
C) import { fs } from 'node';
D) const fs = import('fs');

70. Which Node.js framework is commonly used for building web applications and APIs?
A) Django
B) Express
C) Flask
D) Spring Boot

Printed Pages: 17 **University Roll No.....**

End-Term Examination, Odd Semester 2023-24

B.Tech. CSE, II : Year & III: Semester

BCSE-0902: Smart Industrial Connectivity

Time: 1.5-hour **Maximum Marks: 50**

SECTION A

Note: Attempt All Questions. **30 x 1 =30 Marks**

Read the following code and give the answer from Question 1 to 5

```
#define ledPin 7
int state = 0;
void setup() {
 pinMode(ledPin, OUTPUT);
 digitalWrite(ledPin, LOW);
 Serial.begin(38400); }
void loop() {
 if(Serial.available() > 0){
  state = Serial.read(); }
if (state == '1') {
 digitalWrite(ledPin, HIGH);
 Serial.println("LED: ON");
 state = 0; }
}
```

1. What is the purpose of the if(Serial.available() > 0) { statement in the loop() function?

A. To check if there is data available to be read from the serial port
B. To set pin 7 on the Arduino board to a HIGH state
C. To print the message "Serial: Data Received" to the serial monitor
D. To read the value of pin 7 on the Arduino board

2. What is the purpose of the state = Serial.read(); statement in the loop() function?

A. To read the data from the serial port and store it in the variable state
B. To set pin 7 on the Arduino board to a LOW state
C. To print the message "Serial: Data Read" to the serial monitor
D. To read the value of pin 7 on the Arduino board

3. What is the purpose of the Serial.begin(38400); statement in the setup() function?

A. To define a constant variable named Serial with the value 38400
B. To declare a variable named Serial of type int and initialize it to 38400
C. To set the baud rate of the serial communication to 38400
D. To print the message "Serial: ON" to the serial monitor

4. What is the purpose of the if (state == '0') { statement in the loop() function?

A. To check if the value of the variable state is equal to the character '0'
B. To set pin 7 on the Arduino board to a HIGH state
C. To print the message "LED: OFF" to the serial monitor
D. To read the value of pin 7 on the Arduino board

5. What is the purpose of the #define ledPin 7 statement?

A. To define a constant variable named ledPin with the value 7
B. To declare a variable named ledPin of type int and initialize it to 7
C. To set pin 7 on the Arduino board as an output pin
D. To read the value of pin 7 on the Arduino board

6. Which method is used to calculate basic statistics on a DataFrame, excluding NaN values?

A. stats()
B. describe()
C. summary()
D. basic_stats()

7. What function is used to remove duplicate rows from a DataFrame df?

A. remove_duplicates()
B. drop_duplicates()
C. delete_duplicates()
D. df.remove_duplicates()

8. In Pandas, what is the purpose of the loc attribute?

A. Location based Indexing
B. Logical based Indexing
C. Integer based Indexing
D. Column based Indexing

9. Which of the following is NOT a valid data type in MySQL?

A. INT
B. VARCHAR
C. BOOLEAN
D. TIMESTAMP

10.How can you concatenate two DataFrames, df1 and df2, vertically (stacking one on top of the other)?

A. f1.concatenate(df2, axis=0)

B. pd.concat([df1, df2], axis=0)

C. df1.append(df2)

D. merge(df1, df2, how='vertical')-B

11.Which of the following is not a typical data quality issue?

A. Completeness
B. Consistency
C. Complexity
D. Accuracy

12.What is the purpose of the PRIMARY KEY constraint?

A. To ensure that each row in a table has a unique identifier
B. To prevent duplicate values from being stored in a table
C. To enforce referential integrity between tables
D. To optimize the performance of data retrieval

13. Which of the following SQL statements is used to insert a new row into a table?

A. SELECT
B. INSERT
C. UPDATE
D. DELETE

14. What is the purpose of the WHERE clause in an SQL statement?

A. To specify which rows to retrieve from a table
B. To filter the results of an SQL query
C. To join two or more tables together
D. To group rows by a specific column or columns

15. Which of the following SQL statements is used to join two tables together?

A. SELECT
B. INSERT
C. UPDATE
D. DELETE

16. What is the purpose of the GROUP BY clause in an SQL statement?

A. To group rows by a specific column or columns
B. To calculate aggregate functions over a set of rows
C. To sort the results of an SQL query
D. To limit the number of rows returned by an SQL query

17. What is the purpose of the having clause in an SQL statement?

A. To apply filters to grouped rows
B. To calculate aggregate functions over grouped rows
C. To sort grouped rows
D. To limit the number of grouped rows returned

18. What is the purpose of the ORDER BY clause in an SQL statement?

A. To sort the results of an SQL query
B. To group rows by a specific column or columns
C. To limit the number of rows returned
D. To filter the results of an SQL query

19. What are the main characteristics of the CAN bus?

A. High speed, data integrity, and bus arbitration
B. Low latency, reliability, and flexibility
C. Wide range, high bandwidth, and low power consumption
D. Simple implementation, low cost, and wide adoption

20. How do you connect the MCP2515 to a microcontroller?

A. Serial peripheral interface (SPI)
B. Inter-Integrated Circuit (I2C) bus
C. Universal asynchronous receiver-transmitter (UART)
D. Parallel data bus

21. What is the CAN bus message frame format?

A. The structure of CAN bus data packets, including message ID, data length, and error detection codes
B. The prioritization scheme for CAN bus messages
C. The method of encoding CAN bus messages into binary data
D. The technique of decoding CAN bus messages from binary data

22. What is a Zigbee coordinator?

A. A special device that initiates and manages a Zigbee network
B. A device that receives and transmits Zigbee messages
C. A device that provides routing for Zigbee messages
D. A device that manages network security

23. What is a Zigbee end device?

A. A device that communicates directly with a Zigbee coordinator
B. A device that relays Zigbee messages to other devices in the network
C. A device that provides routing for Zigbee messages
D. A special device that initiates and manages a Zigbee network

24. What is Zigbee mesh networking?

A. A type of network topology in which devices can relay messages for each other, extending the range of the network
B. A method of connecting Zigbee devices to a central hub
C. A technique for encrypting Zigbee messages
D. A way to configure Zigbee devices for different applications

25. What is a common attack vector for IoT devices?

A. Botnets
B. Denial-of-service attacks
C. Man-in-the-middle attacks
D. All of the above

26. What is the role of risk assessment in IoT security management?

A. To identify and prioritize IoT security risks
B. To implement security controls to mitigate risks
C. To monitor and assess the effectiveness of security controls
D. To respond to security incidents

27. What is the purpose of a security gateway in IoT deployments?

A. To enforce access control policies between devices and networks
B. To inspect and filter traffic between devices and networks
C. To provide centralized management of security devices
D. To monitor and log security events

28. What is the Zigbee stack?

A. A collection of software protocols that enable Zigbee devices to communicate with each other
B. A hardware component that provides Zigbee connectivity for microcontrollers
C. A network interface card that connects Zigbee devices to a network
D. A software library that implements Zigbee protocols

29. How does Kepware connect to devices in a ThingWorx environment?

A. Through direct network connection using device IP address
B. Through a data acquisition gateway
C. Through pre-defined drivers for specific protocols
D. By manual configuration of communication parameters

30. What is NOT a primary advantage of using Kepware with ThingWorx?

A. Reduced development time for industrial IoT applications
B. Increased compatibility with diverse industrial protocols
C. Enhanced data security through Kepware software
D. Improved device management within ThingWorx platform

SECTION B

Note: Attempt All Questions. **40 x 0.5 =20 Marks**

31. What are the different data types in MySQL?
    A. Integer, float, character, date, and time
    B. Boolean, string, and decimal
    C. Text, blob, and medium blob
    D. All of the above

32. What are the different data constraints in MySQL?
    A. Primary key, foreign key, and unique key
    B. Not null constraint, default constraint, and check constraint
    C. Auto-increment constraint, index constraint, and partition constraint
    D. All of the above

33. What is the purpose of an index in MySQL?
    A. To improve the performance of data retrieval by quickly locating rows in a table
    B. To enforce referential integrity between tables
    C. To prevent duplicate values from being stored in a table
    D. To optimize the performance of data modification

34. What is the purpose of a foreign key constraint?
    A. To enforce referential integrity between two tables by linking related rows
    B. To prevent orphan records from being created in the child table
    C. To ensure data accuracy and consistency across multiple tables (
    D. To improve the performance of data retrieval and manipulation

35. What is the difference between a SELECT statement and an UPDATE statement?
   A. SELECT statement is used to retrieve data from a table, while UPDATE statement is used to modify data in a table.
   B. SELECT statement uses the WHERE clause to filter data, while UPDATE statement uses the SET clause to modify data.
   C. SELECT statement returns a result set, while UPDATE statement returns an integer indicating the number of rows affected.
   D. All of the above

36. What is the difference between a prepared statement and an ordinary SQL statement?
   A. A prepared statement is precompiled and stored by the database server, improving execution efficiency, while an ordinary SQL statement is compiled and executed each time it is run.
   B. A prepared statement allows for parameter binding, reducing the risk of SQL injection attacks, while an ordinary SQL statement directly embeds parameters into the query string.
   C. A prepared statement is more flexible for handling dynamic data, while an ordinary SQL statement is simpler for basic queries.
   D. A prepared statement is primarily used for bulk data operations, while an ordinary SQL statement is used for a wider range of query

37. Which wireless communication protocol is commonly used for industrial applications due to its reliability and low power consumption?
   A. Bluetooth
   B. Zigbee
   C. Wi-Fi
   D. NFC

38. Which of the following is a key feature of edge computing in industrial connectivity?
   A. Centralized processing
   B. Low-latency data processing
   C. Dependency on cloud servers
   D. Slow response time

39. Which communication protocol is commonly used for real-time control in industrial automation?
   A. HTTP
   B. SNMP
   C. Modbus
   D. SMTP

40. Which communication standard is commonly used for wireless sensor networks in industrial applications?
   A. Bluetooth Low Energy (BLE)
   B. Infrared Data Association (IrDA)
   C. Long-Term Evolution (LTE)
   D. Wireless Fidelity (Wi-Fi)

41. In the context of industrial cybersecurity, what does the term "firewall" refer to??
   A. Physical barrier in a manufacturing facility
   B. Security software that monitors and controls network traffic
   C. Emergency shutdown system
   D. A device for measuring temperature in industrial processes

42. What is the function to create a Numpy array of ones?
   A. numpy.ones()
   B. numpy.one_array()

C. numpy.create_ones()
D. numpy.full_array()

43: Which of the following attributes of DataFrame is used to display all the values from DataFrame?
A. values
B. Values
C. val
D. Val

44. The data label associated with a particular value of Series is called ?
A. Data value
B. Index
C. Value
D. None of Above

45. How can you get the bottom n rows of a DataFrame?
A. df.tail(n)
B. df.last(n)
C. df.bottom(n)
D. df.select(n, from_end=True)

46. How can you remove duplicate rows from a DataFrame?
A. df.unique()
B. df.remove_duplicates()
C. df.filter_duplicates()
D. df.drop_duplicates()

47. Which of the following operations can be performed on a Pandas Series?
A. Filtering
B. Arithmetic operations
C. Sorting
D. All of the above

48. How can you select multiple columns in a data frame?
A. df['column1', 'column2']
B. df[['column1', 'column2']]
C. df[column1, column2]
D. df.column1, column2

49. How do you read a CSV file in pandas?
A. pandas.read_csv()
B. pandas.load_csv()
C. pandas.import_csv()
D. pandas.fetch_csv()

50. What are Pandas?
A. A programming language
B. A data analysis library in Python
C. A machine learning algorithm
D. A statistical software

51. If a dimension is given as ___ in reshaping operation, the other dimensions are automatically calculated?
A. Zero
B. One
C. Negative One
D. Infinite

52. Numpy package able to do the fast operations on arrays?
A. False
B. True
C. None of Above
D. Both a and b

53. Which of the following is the method to change the data type of an array?
A. dtype
B. datatype
C. Astype
D. astype

54. Which of the following is not an argument of the arrange method?
A. start
B. stop
C. end
D. add

55. Which of the method is used to check the dimension of an array
A. Shape
B. dim
C. shape
D. Dim

56. What is the method to find the sum of elements in a Numpy array?
A. array.sum()
B. numpy.sum()
C. array.add()
D. numpy.total()

57. What is the function to create a Numpy array?
A. numpy.array()
B. numpy.asarray()
C. numpy.create_array()
D. numpy.make_array()

58. What is the function to create a Numpy array of zeros?
A. numpy.zeros_array()
B. numpy.zeros()
C. numpy.empty_array()
D. numpy.create_zeros()

59. What is the method to find the data type of a Numpy array?
A. ndarray.item_type
B. ndarray.type
C. ndarray.data_type
D. ndarray.dtype

60. In the given code dataFrame df1 has ___rows and ____columns?

```
Import pandas as pd
Dict = {"Name":["Amit" , "Raj" , "Dev"], "RollNo":[1,2,3]}
Df1 = pd.dataFrame(dict)
```

A. 3,3
B. 2,3
C. 3,2
D. None of Above

61. Which of the following commands is used to create a series ?
A. createSeries()
B. series()
C. create_series()
D. Series()

62. A _____ is a two dimensional labeled data structure ?
A. DataFrame
B. series
C. List

D. Tuple

63. A_____ is a one-dimensional Array?
A. DataFrame
B. Series
C. Both of Above
D. None of Above

64. Which of the following is not a type of data analytics?
A. Descriptive
B. Subscripted
C. Predictive
D. prescriptive

65. Which of the following commands is used to install pandas?
A. pip install pandas
B. install pandas
C. pip pandas
D. None of Above

66. What is the method to fill missing values in a pandas DataFrame or Series?
A. update()
B. replace()
C. fillna()
D. impute()

67. What is the method to find unique values in a pandas DataFrame or Series?
A. unique()
B. nunique()
C. count_unique()
D. get_unique()

68. How can you get the top n rows of a DataFrame?
   A. df.top(n)
   B. df.first(n)
   C. df.head(n)
   D. df.select(n)

69. How can you concatenate two or more DataFrames?
   A. df.concat([df1, df2])
   B. df.append([df1, df2])
   C. df.merge([df1, df2])
   D. df.combine([df1, df2])

70. Which layer of the OSI model does CAN primarily operate in?
   A. Physical Layer
   B. Data Link Layer
   C. Network Layer
   D. Transport Layer

Course Name: Statistics and Complex Analysis

Course Outcome:

CO 1: Compute moments and use them to visualize symmetry and peakedness of the data

CO 2: Fit a straight line on the given data by the method of Least squares

CO 3: Measure the extent of correlation between two variables

CO 4: Predict the value of a variable by finding regression lines

CO 5: Apply discrete probability distributions to the real-life data problems

CO 6: Apply Chi-square test to test a statistical hypothesis in real-life data analysis

CO 7: Know the concepts of analytic and harmonic functions and their applications

CO 8: Solve problems involving complex integration

CO 9: Find radius and region of convergence of a power series

CO 10: Calculate residue and apply Cauchy's residue theorem in solving real integrals

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End Term Examination, Odd Semester 2023-24

B. Tech. (CS, CS – Spec., EC, EC- Spec., EE, CE), II-Year, III-Semester

BMAS 0106: Statistics and Complex Analysis

Time: 3 Hours **Maximum Marks: 50**

Instruction for students:

1. Attempt all questions.
2. Solution must be brief, complete and to-the-point.

Section – A

Attempt All Questions. 4 X 5 = 20 Marks

<table>
<tr><th>S.No.</th><th>Detail of Question</th><th>Marks</th><th>CO</th><th>BL</th><th>KL</th></tr>
<tr><td>1</td><td>What do you mean by correlation? Give some real life examples of correlation between two variables. Calculate Karl Pearson's coefficient of correlation between x and y from the following data:
<table><tr><td>x</td><td>1</td><td>3</td><td>5</td><td>7</td><td>8</td><td>10</td></tr><tr><td>y</td><td>8</td><td>12</td><td>15</td><td>17</td><td>18</td><td>20</td></tr></table></td><td>4</td><td>3</td><td>U</td><td>F</td></tr>
<tr><td>2</td><td>What are regression lines? Write their equations. Find the regression line of y on x from the following data:
<table><tr><td>x</td><td>1</td><td>3</td><td>5</td><td>7</td><td>9</td></tr><tr><td>y</td><td>15</td><td>18</td><td>21</td><td>23</td><td>22</td></tr></table></td><td>4</td><td>4</td><td>U</td><td>F</td></tr>
</table>

3	In a sample analysis of examination results of 500 students, it is found that 220 students failed, 170 secured a third class, 90 secured a second class and the rest, a first class. Do these figures support the general belief that above categories are in the ratio 4: 3: 2: 1 respectively? [Given that the tabular value of χ^2 for degree of freedom 3 at 5 % level of significance is 7.81.]	4	6	A	C
4	Define 'Analytic Functions'. Show that the function $f(z)=e^{z}$ is analytic everywhere in finite complex plane.	4	7	U	C
5	Show that for the function, $f(z)=\begin{cases}\dfrac{x^2y^5(x+iy)}{x^4+y^{10}}, & z\neq 0\\ 0, & z=0\end{cases}$ the Cauchy-Riemann equations are satisfied at the origin but derivative of $f(z)$ does not exist at origin.	4	7.	An	P

Section – B

*Attempt **All** Questions.* 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Determine an analytic function $f(z)$ in terms of z whose real part is $e^{-x}(x\sin y - y\cos y)$.	3	7	E	C
7	Prove that the functions, $u=\frac{1}{2}\log(x^2+y^2)$ and $v=tan^{-1}\frac{y}{x}$ are harmonic conjugates of each other.	3	7	U	C

8	Evaluate: $\int_0^{2+i} (\bar{z})^2 \, dz$, along the real axis from $z = 0$ to $z = 2$ and then along the line parallel to y-axis from $z = 2$ to $z = 2 + i$.	3	8	A	P
9	Find the radius of convergence of power series $\sum_{n=0}^{\infty} \left(\frac{z^n}{2^n + 1} \right)$	3	9	U	C
10	Expand the function $f(z) = \frac{1}{(z+1)(z+3)}$ in the following regions: (a) $1 < \lvert z+1 \rvert < 2$ and (b) $1 < \lvert z \rvert < 3$.	3	10	An	P

Section – C

*Attempt **All** Questions.* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	Determine poles and residue at each pole of the function $f(z) = \frac{z^2}{(z-1)^2(z+2)}$. Hence evaluate $\int_C \frac{z^2}{(z-1)^2(z+2)} \, dz$, where $C \equiv \lvert z \rvert = 3$.	5	10	An	P
12	Discuss the various types of singularities in a function of complex variable and give suitable examples.	5	7	U	C

13	**Case study:** Cauchy Integral theorem states that "if a function F (z) is continuous at each point within and on a simple closed curve C then integral of that function along the curve C remains zero". However, Goursat later proved that the assumption of continuity of derivative of f (z) is unnecessary and Cauchy's theorem holds if f (z) is analytic within and on the curve C. It was made applicable to multi-connected regions by converting multi-connected regions to simply connected regions. Subsequently, Cauchy used the concept of residues to propose the Cauchy residue theorem which states that: $$\oint_c f(z)\,dz = 2\pi i\,(R_1 + R_2 + R_3 + \cdots)$$ where R_1, R_2, R_3,... are the residues of f (z) at the poles lying within the curve C. **Based on the above case study, evaluate the following complex integral,** $$I = \oint_c \frac{dz}{(z^2+1)(z^2-4)}$$ where, (a) $C \equiv \lvert z \rvert = 0.5$ (b) $C \equiv \lvert z \rvert = 2$ (c) $C \equiv \lvert z - i \rvert = 1$ (d) $C \equiv \lvert z + i \rvert = 1$ (e) $C \equiv \lvert z \rvert = 5$	5	8,10	E	P

Course Name: English For Professional Purposes I

Course Outcome

CO1- create structurally effective, grammatically accurate and contextually appropriate written and spoken pieces for communication both independently and collaboratively.

CO2-display competence in all aspects of language learning not limited to listening, speaking, reading and writing.

CO3-develop independent perspectives and points of view and express them in a cohesive and meaningful way.

CO4-produce a repertoire of register-specific and academic vocabulary appropriate in both oral and written communication.

CO5-use language as an effective tool of communication for various contexts in both academic and professional setting.

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End Term Examination, Odd Semester 2023-24

B.Tech II Year, III Semester (All Branches) & MCA I Year, I Semester

BELH 1003 English for Professional Purposes I

Time: 3 Hours **Maximum Marks: 50**

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Mark	CO	BL	KL
1	i-Identify the grammatical function of the word fast in each of the following sentences 1- By the time I got home, the children were already fast asleep. A. Noun B. Verb C. Adverb D. Preposition 2- Some prisoners began a fast to protest against the appalling conditions A. Verb	4	1	A	F

	B. Adjective C. Noun D. Interjection i-Kelly has great confidence _____ her children's ability. Choose the correct preposition. A. on B. at C. for D. in iii. Unfortunately, many online courses do not provide students _____ the basic skills they need. A. with B. for C. on D. in iv. They hope to can arrive before the start of the conference. Identify the unnecessary word in the sentence and its part of speech. A. can- linking verb B. to- infinitive C. the (second) – Definite article D. can - Auxiliary Verb v. J: I don't have any money. K: I don't have no money. L: I don't have money. The grammatically correct sentences are: A. J and K B. K and L C. J and L D. J, K and L vi. Leap year falls in _____ forth year. A. Each B. Every C. One				

	D. Any vii. He showed _____ concern for his nephew. A. the little B. a few C. little D. a little viii. Whether I shall turn out to be the hero of my own life, or whether that station will be held by anyone else, _____ pages must show. A. these B. this C. none D. third ix. _____truth must be spoken. A. a B. the C. no article D. Both A and B. x.In which year were both these articles circulated to all the students? Identify the determiners. A. to, all, these, both. B. both, the, all, in. C. to, year, all, the. D. these, the, both, all xi. Linear communication is _____ one-way process conducted in ____ straight line towards ____ main point. Fill in the blanks with correct determiners. A. a, the, the B. the, a, a C. a, the, a D. a, a, the xii. Most men and women will grow up to love _____				

servitude and will never dream of revolution.

Fill in the blank with the correct determiner.

A. our
B. there
C. their
D. Many

xiii-The manager said to me, "Complete it by tomorrow."

A. The manager requested me to complete it by tomorrow.
B. The manager instructed me to complete it by the next day.
C. I was ordered by the manager to complete the work by the next day.
D. The manager ordered me to complete the work by the next day.

xiv- J) Today it was first time we play such games.
k) Bacteria can kill you into a few days.
L) I am leaving a call at 10:00 AM.

The grammatically correct sentence is:

A. J and K
B. K and L
C. J and L
D. None of the above

xv- J) I will buy some mangoes.
K) I will not buy some mangoes.
L) A little knowledge is a dangerous thing.
M) A darkest cloud has a silver lining.

Which of the following sentences have the correct use of determiners?

A. J and L only
B. K, L and M are correct
C. Only J is correct
D. Only M is correct

xvi- He gave a talk _____ the causes of Dengue.

A. on

	B. for C. at D. of				
2	i He is an *absolutely honest* politician. A. Noun Phrase B. Prepositional Phrase C. Adjective Phrase D. Adverb Phrase ii.*In the realm of quantum mechanics*, researchers delve into the mysteries of subatomic particles. A. Prepositional Phrase B. Verb Phrase C. Adverb Phrase D. Independent Clause iii. The arrival *of the train* has been delayed. A. Adverb Phrase B. Prepositional Phrase C. Noun Phrase D. Verb Phrase iv. *All my sons* are good football players. A. Noun Phrase B. Adjective Phrase C. Adverb Phrase D. Verb Phrase v. It *has been raining* continuously for the last seven days. A. Verb Phrase B. Adverb Phrase C. Adjective Phrase D. Noun Phrase vi. The men in the red scarves dance *quite merrily* in the sun. A. Verb Phrase B. Noun Phrase C. Adverb Phrase D. Adjective Phrase vii.The people look solemn, yet they go to work every day. A. Independent Clause followed by a Dependent Clause. B. Dependent Clause followed by an Independent Clause. C. 2 Independent Clauses	4	1	U	F

	D. 2 Dependent Clauses viii. *Although the train arrived on time*, we still missed it. A. Dependent Clause B. Independent Clause C. Prepositional Phrase D. Verb Phrase ix. Jane's reputation as a/an _____ grew as she consistently prioritized her own needs over those of her colleagues. A. egoist B. versatile C. flamboyant D. selfish x. Before responding, he took a moment to _____, the impact of his words on the conversation. A. contemplate B. introspect C. adapt D. accustom xi. When choosing mutual funds, individuals often prefer those with a ____ history of reliable returns and performance. A. bankrupt B. bankable C. prominent D. amateur xii.The old computer system became _____ after the company upgraded to the latest technology. A. famous B. redundant C. obsolete D. obliterate xiii. The Indian Army _____ INS Viraat in the year 2017. A. decommissioned B. simulate C. reinstitute D. rehabilitate xiv. The math _____ effortlessly solved complex problems that left even seasoned mathematicians impressed. A. novice B. servile				

	C. gifted D. prodigy xv. The jury decided on the capital punishment _____ A. impromptu B. solely C. unanimously D. servile xvi. The _____ of the situation was that the more she tried to control time, the more it seemed to slip away from her grasp. A. paradox B. agenda C. veteran D. fanciful				
3	i. Rewrite the following *para* to make it coherent. "Limited investment in the housing sector makes it practically impossible to allocate sufficient resources for urban dwellers' housing needs. A high rate of urban population growth has increased the country's needs for housing. A small group of city officials has laid out a new plan to combat the crisis. A solution to the housing-shortage problem is a vital policy issue here. The housing problem has grown in the last twenty years." ii. Fill in the blanks with the most suitable transitions from the list. Use each transition only once: in spite of, thanks to, although, similar to, like, also, in addition to, whereas, compared to, A._____ we have scientific evidence that women should not smoke when they are pregnant, many future mothers ignore this danger. These mothers endanger the physical health of their babies. B-_____, they could be contributing to even more problems in the future. C- _____ the known problems, recent studies have linked mothers' smoking during their pregnancy with sons who go on to become criminals. Researchers in Finland followed over 5000 males and their mothers from the time the mothers were six months pregnant until the young men	2+2	2	A	C

	were 28 years old. D- _____ sons whose mothers did not smoke during pregnancy, those young men whose birth mothers did smoke were more than twice as likely to commit criminal acts. The researchers theorized that smoking during pregnancy may alter the chemical balance in the baby's brain.				
4	Read the following passages and answer the questions that follow: About once in every seven years, the ocean surface off the coast of Peru warms up. This cuts the normal enriching of nutrient-rich cold water. Plankton production is drastically reduced. This phenomenon, known as El Nino (the child) because it starts during the Christmas season, usually lasts for up to a year. Occasionally, it goes on for long. Three times this century, it has persisted into a third year. The latest of these prolonged episodes have been blamed for much more than its effect on the Peruvian fishery. It certainly seems to have played a part in droughts in normally humid Indonesia as well as those that brought catastrophic fires to the outskirts of Sydney. It has been blamed for storms and landslides in coastal regions of Peru and Equator and is associated with drought in north-eastern Brazil. Its influence may stretch as far as western Europe where the recent winter brought heavy rainfall and flooding. El Nino is not the only ocean phenomenon to affect the weather. To understand how this might be done, it is useful to think about how the weather is forecast and about what makes it predictable. Weather forecasting uses two types of techniques, both of which rely on observations of what is going on at and shortly before the time of the forecast. To supply this information, national authorities operate extensive land, sea and air based observation networks. These are coordinated under the United Nations World Meteorological Organisation. There is an effective and almost instantaneous worldwide exchange of the information gathered. The first, classical forecasting method compares the set of observations with experience and bases the forecast on what has happened in the past from a similar starting point. The second, only possible since the advent of fast computers, applies the laws of physics through a numerical model of the atmosphere to	4	2	U	F

predict how the weather will develop from the observed starting point. This method is the basis of modern weather forecasting. But neither method will produce useful predictions for more than a few days ahead. This is because the behaviour of the atmosphere is governed in the medium term by energy exchanges with the land and ocean. The latter is much more important as the sea surface stores energy which it exchanges with the atmosphere in a surface layer 100 m or so thick. The energy storage capacity of the layer is many times greater than that of the whole atmosphere. This had prompted the suggestion that longer term forecasting might be practicable if a coupled computer model of the atmosphere and ocean could be constructed.

A great deal of research has been put into this with some success but progress has been limited by two factors. The first is that coupled ocean-atmosphere models require enormously greater computing capacity than is provided even by the super computers used in weather forecasting. The second is that not enough is known about the state of the ocean at any given time - there is no global observation network as there is for the atmosphere–or about the processes that govern the interactions. When these limitations have been overcome, it seems likely that coupled models will permit prediction of such climatological factors as frequency and intensity of rainfall for seasons and perhaps years ahead.

In the meantime, climatologists have begun to be able to predict the onset and consequences of phenomena such as El Nino using techniques like those used in the classical weather forecasting method. These examine the condition of the ocean at a given instant and, by comparison with past experience, attempt to predict in a statistical way how the ocean or the atmosphere—and hence the climate—are likely to behave for the next few months. As with the weather forecasts, this depends critically on the existence of a bank of past experience, in this case, particularly of sea-surface temperatures. A key contribution to this, recently completed by the UK Meteorological Office, has been the analysis of many millions of sea-surface temperatures. This has led to the publication of the first globally complete monthly fields of sea-surface temperature from 1871 to the present day. This

type of information may be used to predict events such as El Nino.
Various scientific groups have tried to do so with varying degrees of success. In the meantime, Meteorological Office scientists have also compared the temperature fields statistically with climatological factors. They have shown that, particularly in the tropics, there are significant correlations between sea-surface temperature anomalies and climate statistics. This does not necessarily mean that one causes the other, though some degree of direct linkage seems likely. But it does open up the possibility of predicting short-term climate fluctuations.
The UK Meteorological Office has approached this by a rigorous comparison between rainfall statistics in the Nordeste area of Brazil, whose crops can be seriously affected by drought, with contemporaneous sea-surface temperatures world-wide. This has revealed significant links, verified over the period 1901-85, between rainfall and sea-surface temperatures in the north and south tropical Atlantic and the western tropical Pacific, the area most strongly affected by El Nino.

1. Which of the following is NOT TRUE about 'El Nino'?
(1) It is the most important ocean phenomenon to affect a region's weather pattern.
(2) There seems to be a statistical link between sea-surface temperatures and the occurrence of El Nino.
(3) The consequences of El Nino can vary with the geographical position of the affected area.
(4) As El Nino results in higher oceanic temperatures, the number of fish comes down due to lower plankton production.
2. Which of the following data will be helpful in the prediction of El Nino?
(1) Daily temperatures recorded in coastal areas.
(2) Daily atmospheric pressure levels for one previous year.
(3) A data bank of sea-surface temperatures.
(4) All of the above
3. The frequency of occurrence of El Nino, approximately, is
(1) once every year
(2) about three times in a century
(3) once in three years

	(4) once every seven years 4. The passage talks about (1) the factors that can trigger phenomena like El Nino. (2) the advances being made in the field of meteorological predictions through a greater study of ocean-atmosphere interactions. (3) phenomenon like El Nino that affect the planet's climate. (4) the predisposing factors that result in El Nino and the ways to construct them.				
5	Assuming yourself participating in a debate write four arguments in favour or against on one of the following topics- i-Robots increase people's quality of life. ii-Modern parenting has made kids impolite and less developed.	4	3	E	C

Section – B

Attempt All Questions 3 X 5 = 15

Marks

No.	Detail of Question	Marks	CO	BL	KL
6	i-What tips will you give to your friend about proper use of eye contact in Presentation? Choose the correct option- 1. Deciding it was time to sit down and begin a novel, the would-be writer spent the first several hours deciding upon a suitable ..... A.extempore B.nom de plume C.ad hoc D.vis à vis 2. Central High School is my ..... A. au revoir B. vide C. sangfroid D. alma mater 3. He has a ..... of knowledge when it comes to human psychology.	1+2	5,4	U,R	F

	A. plethora B. ad hoc C. pro bono D. phobia 4. The restaurant had a delightful ..... A. gratis B. status quo C. ambiance D. gourmet				
7	i. Having encountered several challenges, they _____ unchartered territories now. a. hated to explore b. explore c. are exploring d. hate Choose the most suitable option: A. only a B. only d C. Both c and d D. Both b and d ii. The project _____ undergone significant modifications. Use the most suitable primary auxiliary verb: A. is B. is doing C. has D. was iii. He _____ have completed the task by now. Choose the least suitable modal auxiliary verb: A. ought to B. must C. should D. can iv. _____ a marathon requires months of dedicated training. Use a non-finite verb as a noun: A. Winning	3	I	A	F

	B. Run for C. Participating D. Training v. The decision _____ these changes was not taken lightly. Choose the most appropriate option: A. was taken to implement B. is to implement C. to implement D. implement vi. I need to go to the doctor as I think that I might have a chronic disease. Which of the following verbs is NOT used in the sentence given below? A. primary auxiliary verb B. main verb C. semi modal auxiliary verb D. non-finite verb				
8	**Read the following passage. Identify sentence type-** i-These interfaces accept requests written in English or some other language and attempt to understand them. ii. A natural language interface usually has its own schema, which is similar to the database conceptual schema, as well as a dictionary of important words. iii. The natural language interface refers to the words in its schema, as well as to the set of standard words in its dictionary, to interpret the request. iv. If the interpretation is successful, the interface generates a high-level query corresponding to the natural language request and submits it to the DBMS for processing; otherwise, a dialogue is started with the user to clarify the request. v. The capabilities of natural language interfaces have not advanced rapidly. vi. Today, we see search engines that accept strings of natural language (like English or Spanish) words and match them with documents at specific	3	1	U	F

	sites (for local search engines) or Web pages on the Web at large (for engines like Google or Ask). i A. Simple B. Complex C. Negative D. Compound ii. A. Simple, complex B. Complex, compound C. Simple D. Compound, complex iii. A. Simple, complex B. Complex, compound C. Simple D. Compound iv A. Simple, complex B. Complex, compound C. Simple D. Compound, complex v. A. Simple B. Complex C. Complex, Compound D. Compound vi A. Simple, complex B. Complex, compound C. Simple D. Compound, complex				
9	Read the following passage and answer the questions that follow- In the forsaken outskirts of the bustling city, where dim lights flicker and children fear to tread at night, a mysterious figure appeared. It was the detective known for solving	3	1	A	F

	perplexing cases. The police department was instructed to carefully examine the evidence and submit a preliminary investigation report. The detective could help piece together fragments of a puzzle that seemed unsolvable. The witnesses whispered about the rumours surrounding the young man's demise as the detective hurriedly typed something on her phone. Having hurt her foot in the day, the detective struggled to jump over the stretched wires. The pine trees stood alongside the road like students lined up in a parade but they appeared formidable, not fragile. A suffocating smell filled the air as the night wore on and the city held its breath awaiting the revelation of the truth. In the above paragraph, identify the following: i. Two past participles used as adjectives ii. a. One present participle used as a verb, b. One perfect participle iii. a. One linking verb, b. One modal auxiliary verb iv. Two transitive verbs along with their objects v. Two intransitive verbs vi. one bare infinitive and one split infinitive				
10	Frame sentences as directed- i- Inquire about the objective of your project. ii- Show surprise on receiving an award. iii-Share any experience in a complex sentence. iv-Instruct your friend the way to the hostel in simple sentence. v-Congratulate your friend on his placement. vi-Write a declarative sentence about GLA University.	3	2	C	F

Section – C

Attempt All Questions 5 X 3 = 15

Marks

No.	Detail of Question	Marks	CO	BL	KL
11	The sentences (labeled 1, 2, 3, 4, 5) given in this question, when properly sequenced, form a coherent paragraph. Each sentence is labeled with a number. Decide on the proper order for the sentences and key in this sequence of five numbers as your answer. i- 1. The implications of retelling of Indian stories, hence, takes on new meaning in a modern India. 2. The stories we tell reflect the world around us. 3. We cannot help but retell the stories that we value — after all, they are never quite right for us — in our time. 4. And even if we manage to get them quite right, they are only right for us — other people living around us will have different reasons for telling similar stories. 5. As soon as we capture a story, the world we were trying to capture has changed. A. 25341 B. 32145 C. 12345 D. 42315 ii- 1. People with dyslexia have difficulty with print-reading, and people with autism spectrum disorder have difficulty with mind-reading. 2. An example of a lost cognitive instinct is mind-reading: our capacity to think of ourselves and others as having beliefs, desires, thoughts and feelings. 3. Mind-reading looks increasingly like literacy, a skill we know for sure is not in our genes, since scripts have been around for only 5,000-6,000 years. 4. Print-reading, like mind-reading varies across cultures, depends heavily on certain parts of the brain, and is subject to developmental disorders. A. 3214 B. 2341 C. 4321 D. 1234 iii- 1. If you've seen a little line of text on websites that says something like "customers who bought this also enjoyed that" you have experienced this collaborative filtering firsthand.	5	2	U	C

2. The problem with these algorithms is that they don't take into account a host of nuances and circumstances that might interfere with their accuracy.
3. If you just bought a gardening book for your cousin, you might get a flurry of links to books about gardening, recommended just for you! – the algorithm has no way of knowing you hate gardening and only bought the book as a gift.
4. Collaborative filtering is a mathematical algorithm by which correlations and cooccurrences of behaviors are tracked and then used to make recommendations.

A. 3214 B. 4123 C. 1234 D. 2314

iv-

1. It advocated a conservative approach to antitrust enforcement that espouses faith in efficient markets and voiced suspicion regarding the merits of judicial intervention to correct anticompetitive practices.
2. Many industries have consistently gained market share, the lion's share – without any official concern; the most successful technology companies have grown into veritable titans, on the premise that they advance 'public interest'.
3. That the new anti competitive risks posed by tech giants like Google, Facebook, and Amazon, necessitate new legal solutions could be attributed to the dearth of enforcement actions against monopolies and the few cases challenging mergers in the USA.
4. The criterion of 'consumer welfare standard' and the principle that antitrust law should serve consumer interests and that it should protect competition rather than individual competitors was an antitrust law introduced by, and named after, the 'Chicago school'.

A.1234 B. 4123 C. 3124 D. 4321

v-

1. The woodland's canopy receives most of the sunlight that falls on the trees.
2. Swifts do not confine themselves to woodlands, but hunt wherever there are insects in the air.
3. With their streamlined bodies, swifts are agile flyers, ideally adapted to twisting and turning through the air as they chase flying insects – the creatures that form their staple diet.

	4. Hundreds of thousands of insects fly in the sunshine up above the canopy, some falling prey to swifts and swallows. A.1234 B. 1324 C. 4321 D. 1432				
12	Rewrite the following sentences in minimum possible words so as to make them as brief as possible- i. As of 4 August, students will be selected using processes which have been created in accordance with legal requirements. These processes will be used along with our diversity strategy and all of its relevant policies about student selection. ii. The Passport Office is replacing free of charge until the dates specified any current passport lost, damaged or destroyed as a result of the recent floods. Eligible applicants should contact the Passport Information Service (PIS) on 998 9988 for further information. iii.If an agency or organization breaches the privacy principles, our office may investigate the matter. Individuals can also make a privacy complaint to us about an agency or organisation if they think their information has been mishandled. iv. Aid Agency works to bring immediate assistance to people whose lives have been devastated by natural disasters and conflict. Through our work we provide emergency relief in the initial stages of the disaster, and we also provide continued support to affected communities as they rebuild and rehabilitate towards a sustainable future. v. It is apparent that the tragedy could have been avoided if the company had talked to workers regarding hazards which existed in the workplace.				
13	You have to give a presentation analyzing the data on the topic 'Is Hybrid Work Sustainable?' Prepare a questionnaire for the same. (Minimum 5 questions & personal information detail)	5	5	An	C

Course Name: Operating Systems

Course Outcome

CO1- Understand the classification of operating system environment.
CO2- Understand the basic of process management.
CO3- Apply the concept of CPU process scheduling for the given scenarios.
CO4- Illustrate the process synchronization and concurrency process in operating system.
CO5- Analyze the occurrence of deadlock in operating system.
CO6- Describe and analyze the memory management and its allocation policies.
CO7- Understand the concepts of disk scheduling

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End Term Examination, Odd Semester 2023-24
B.Tech. (CSE), AIML, CCV, DA, CSF, IIOT & EC (Minor in CS), Year-II, Semester-III
OPERATING SYSTEMS (BCSC 0004)

Time: 3 Hours **Maximum Marks: 50**

Note: Attempt all Sections

Section – A

*Attempt **All** Questions* 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Describe the differences between symmetric and asymmetric multiprocessing. What are three advantages and one disadvantage of multiprocessor systems?	2+2	1	R	F
2	How operating system manages critical section problem by using semaphore? Differentiate between binary and counting semaphore? Write down the Peterson solution for two process synchronization.	1+1+2	4	U	C
3	An operating system uses the banker's algorithm for deadlock avoidance when managing the allocation of three resource types X, Y and Z to three processes P0, P1 and P2. The table given below presents the current system state. Here, the Allocation matrix shows the current number of resources of each type allocated to each process and the Max matrix shows the maximum number of resources of each type required by each process during its execution.	2+2	5	A	P

		Allocation			Max					
		X	Y	Z	X	Y	Z			
	P0	0	0	1	8	4	3			
	P1	3	2	0	6	2	0			
	P2	2	1	1	3	3	3			
	There are 3 units of type X, 2 units of type Y and 2 units of type Z still available. The system is currently in safe state. Consider the following independent requests for additional resources in the current state- **REQUEST1**: P0 requests 0 units of X, 0 units of Y and 2 units of Z **REQUEST2**: P1 requests 2 units of X, 0 units of Y and 0 units of Z Find out whether REQUEST1 and REQUEST2 can be permitted or not?									
4	Consider the 3 processes, P1, P2 and P3 shown in the table: Process — Arrival time — Time Required P1 — 0 — 5 P2 — 1 — 7 P3 — 3 — 4 Find the completion order and calculate average waiting time of the 3 processes under the policies FCFS and RRS (round robin scheduling with CPU quantum of 2 time units).							2+2	3	A	P
5	Why translation lookaside buffers are necessary? A paging scheme uses a Translation Lookaside buffer (TLB). The effective memory access takes 260 ns and a main memory access takes 100 ns. What is the TLB access time (in ns) if the TLB hit ratio is 80% and there is no page fault?							1+3	6	U	C

Section – B

*Attempt **All** Questions* 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Draw the resource allocation graph and determine deadlock or deadlock free state. P = {P1, P2, P3} R = {R1, R2, R3, R4} E = { P1 →R1, P2→R3, R1→P2, R2→P2, R2→P1, R3→P3, P3→R2} There are ONE, TWO, ONE and THREE instances of R1, R2, R3 and R4 type resources	2+1	5	An	C
2	Consider six memory partitions of size 200 KB, 400 KB, 600 KB, 500 KB, 300 KB and 250 KB. These partitions need to be allocated to four processes of sizes 357 KB, 210 KB, 468 KB and 491 KB in that order. Perform the allocation of processes using- 1. First Fit Algorithm 2. Best Fit Algorithm 3. Worst Fit Algorithm	1+1+1	6	A	P
3	The Logical memory and physical memory space is 32k and 512 words respectively. If Page size is of 64 words, find the logical and physical address. How many pages logical and physical memory can accommodate? Write number of entries in the page table also.	1+1+1	6	E	P
4	A process has been allocated 3 page frames. Assume that none of the pages of the process are available in the memory initially. The process makes the following sequence of page references (reference string): 1, 2, 1, 3, 7, 4, 5, 6, 3, 1 If optimal page replacement policy is used, how many page faults occur for the above reference string?	3	6	E	C
5	A disk with 1000 cylinders, 0 to 999, compute the number of track a disk arm must move to satisfy all the request in the disk queue. Assume the last request served was at track 345 and head is moving towards 0. The queue in the FIFO order is	1.5+1.5	7	A	P

	123, 874, 692, 475, 105, 376 Perform the computation for the following Disk Scheduling Algorithm. a) SSTF b) SCAN				

Section – C

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Discuss Linked and Indexed File allocation methods with its advantages and disadvantages. Which allocation scheme will minimize the amount of space required in the directory structure and why?	3+2	7	U	F
2	Consider the disk system with 100 cylinders. The request to access the cylinders occur in the following sequence. 4, 37, 10,7,19,73,2,15,6,20 Assuming the head is currently at cylinder 50 what is the time taken to satisfy all requests if it takes 1ms to move from one cylinder to adjacent one and Scan and C- Look algorithms are used.	2.5+2.5	7	A	P
3	Assume **"Department of CEA, GLA University"** implemented the new page replacement algorithm in virtual memory and given its name as 'CEAR'. Consider the working strategy of CEAR as following- Each page in memory maintains a count which is incremented if the page is referred and no page fault occurs. If a page fault occurs, the physical page with zero count or smallest count is replaced by new page and if more than one page with zero count or smallest count then it uses FIFO strategy to replace the page. Find the number of page faults using CEAR algorithm for the following reference string (assume three physical frames are available which are initially free) and the Reference String: "A B C D A B E A B C D E B A D"	5	6	C	M

Course Name: Computer Organization

Course Outcome

CO1- Understand the basics of digital computer system.
CO2- Demonstrate the principle of arithmetic operations on unsigned, signed integers and floating point numbers.
CO3- Understand the concepts of combinational and sequential circuits and their applications.
CO4- Understand the CPU architecture and organization.
CO5- Explain the basic concepts of pipelining.
CO6- Design the steps for the execution of the complete instruction for hardwired and micro-programmed control unit.
CO7- Explain the function of memory hierarchy.
CO8- Determine the interface of CPU with input/output devices and their modes of transfer.

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End Term Examination, Odd Semester 2023-24
B. Tech. (CS/AIML/CCV/CSF/DA/IIOT), II Year, III Semester
BCSC 1005 : Computer Organization

Time: 3 Hours **Maximum Marks: 50**

Section – A

*Attempt **All** Questions* 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	Bl.	KL
1	Consider two binary numbers $A_3A_2A_1A_0$ and $B_3B_2B_1B_0$ are stored in two registers A & B based on Parallel-In Parallel-Out mode respectively. Design a combinational logic to subtract two numbers.	4	3	U	C
2	Show how a computer processor evaluates the following arithmetic expression by using reverse polish notation? ((54-20)*((100-3*10)/7)/2)/2+15	4	4	A	P
3	Represent $(175.51)_2$ in single and double precision floating point number.	4	2	A	P
4	Consider a cache consisting of 256 blocks of 16 words each for a total of 4096 words. Also, assume that the main memory is addressable by a 16 bits address and accommodate 12 bits data. How many bits are there in each of Tag, Index, Block, and Word field for 2-way set associative technique? Include a valid bit. What is the size of the cache memory?	4	6	U	P
5	Consider a pipeline having 4 phases with duration 60, 50, 90 and 80 ns. Given latch delay is 10 ns. Calculate-	4	5	U	P

	1. Pipeline cycle time 2. Non-pipeline execution time 3. Speed up ratio 4. Pipeline time for 1000 tasks 5. Sequential time for 1000 tasks 6. Throughput Or In a certain scientific computation, it is necessary to perform the following arithmetic expression. $A_i * \left(\frac{B_i}{C_i}\right) + D_i$ (a) Specify the pipeline configuration to carry out the above task. (b) List the content of all registers in a space-time diagram for 5 tasks.				

Section – B

*Attempt **All** Questions* 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Define (a) Locality of reference (b) Write-back (c) Page fault	3	5	R	F
7	A computer uses RAM chips of 1024x1capacity. (a) How many chips are needed, and how should their address lines be connected to provide a memory capacity of 1024 bytes? (b) How many chips are needed to provide a memory capacity of 16K bytes?	3	7	U	C
8	Define (a) Virtual memory (b) Instruction cycle (c) Interrupt	3	6	R	F
9	Explain different methods of asynchronous data transfer.	3	8	R	F
10	What sequence of micro-operations is performed by a computer system to execute the STA instruction for which two references to the memory are required to compute the memory address?	3	7	U	C

Section – C

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	A virtual memory system has an address space of 8K words, a memory space of 4K words, and page & block sizes are of 1K words. The following page reference changes occur during a given time interval. 3 0 2 1 2 1 3 4 0 1 0 3 2 5 6 0. Determine the four pages that are resident in main memory after each page reference change if the replacement algorithm used is (a) FIFO (b) LRU	5	7	U	P
12	Explain a DMA controller and its mode of data transfer with proper illustration.	5	8	R	F
13	A computer employs RAM chips of 256x8 and ROM chips of 1024x8. The computer system needs 2K bytes of RAM, 4K bytes of ROM, and four interface units each with four registers. A memory-mapped I/O configuration used. The two highest-order bits of the address bus are assigned 00 for RAM, 01 for ROM, and 10 for interface registers. (a) How many RAM and ROM chips are needed? (b) Draw a memory-address map for the system.	5	7	U	P

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End-Term Examination, Odd Semester 2023-24

B.Tech. CSE, II : Year & III: Semester

BCSE-0902: Smart Industrial Connectivity

Time: 1.5-hour **Maximum Marks: 50**

SECTION A

Note: Attempt All Questions. **30 x 1 =30 Marks**

Read the following code and give the answer from Question 1 to 5

```
#define ledPin 7
int state = 0;
void setup() {
 pinMode(ledPin, OUTPUT);
 digitalWrite(ledPin, LOW);
 Serial.begin(38400); }
void loop() {
 if(Serial.available() > 0){
  state = Serial.read(); }
if (state == '1') {
 digitalWrite(ledPin, HIGH);
 Serial.println("LED: ON");
 state = 0; }
}
```

1. What is the purpose of the if(Serial.available() > 0) { statement in the loop() function?

A. To check if there is data available to be read from the serial port
B. To set pin 7 on the Arduino board to a HIGH state
C. To print the message "Serial: Data Received" to the serial monitor
D. To read the value of pin 7 on the Arduino board

2. What is the purpose of the state = Serial.read(); statement in the loop() function?

A. To read the data from the serial port and store it in the variable state
B. To set pin 7 on the Arduino board to a LOW state
C. To print the message "Serial: Data Read" to the serial monitor
D. To read the value of pin 7 on the Arduino board

3. What is the purpose of the Serial.begin(38400); statement in the setup() function?

A. To define a constant variable named Serial with the value 38400
B. To declare a variable named Serial of type int and initialize it to 38400
C. To set the baud rate of the serial communication to 38400
D. To print the message "Serial: ON" to the serial monitor

4. What is the purpose of the if (state == '0') { statement in the loop() function?

A. To check if the value of the variable state is equal to the character '0'
B. To set pin 7 on the Arduino board to a HIGH state
C. To print the message "LED: OFF" to the serial monitor
D. To read the value of pin 7 on the Arduino board

5. What is the purpose of the #define ledPin 7 statement?

A. To define a constant variable named ledPin with the value 7
B. To declare a variable named ledPin of type int and initialize it to 7
C. To set pin 7 on the Arduino board as an output pin
D. To read the value of pin 7 on the Arduino board

6. Which method is used to calculate basic statistics on a DataFrame, excluding NaN values?

A. stats()
B. describe()
C. summary()
D. basic_stats()

7. What function is used to remove duplicate rows from a DataFrame df?

A. remove_duplicates()
B. drop_duplicates()
C. delete_duplicates()
D. df.remove_duplicates()

8. In Pandas, what is the purpose of the loc attribute?

A. Location based Indexing
B. Logical based Indexing
C. Integer based Indexing
D. Column based Indexing

9. Which of the following is NOT a valid data type in MySQL?

A. INT
B. VARCHAR
C. BOOLEAN
D. TIMESTAMP

10.How can you concatenate two DataFrames, df1 and df2, vertically (stacking one on top of the other)?

A. f1.concatenate(df2, axis=0)

B. pd.concat([df1, df2], axis=0)

C. df1.append(df2)

D. merge(df1, df2, how='vertical')-B

11.Which of the following is not a typical data quality issue?

A. Completeness
B. Consistency
C. Complexity
D. Accuracy

12.What is the purpose of the PRIMARY KEY constraint?

A. To ensure that each row in a table has a unique identifier
B. To prevent duplicate values from being stored in a table
C. To enforce referential integrity between tables
D. To optimize the performance of data retrieval

13. Which of the following SQL statements is used to insert a new row into a table?

A. SELECT
B. INSERT
C. UPDATE
D. DELETE

14. What is the purpose of the WHERE clause in an SQL statement?

A. To specify which rows to retrieve from a table
B. To filter the results of an SQL query
C. To join two or more tables together
D. To group rows by a specific column or columns

15. Which of the following SQL statements is used to join two tables together?

A. SELECT
B. INSERT
C. UPDATE
D. DELETE

16. What is the purpose of the GROUP BY clause in an SQL statement?

A. To group rows by a specific column or columns
B. To calculate aggregate functions over a set of rows
C. To sort the results of an SQL query
D. To limit the number of rows returned by an SQL query

17. What is the purpose of the having clause in an SQL statement?

A. To apply filters to grouped rows
B. To calculate aggregate functions over grouped rows
C. To sort grouped rows
D. To limit the number of grouped rows returned

18. What is the purpose of the ORDER BY clause in an SQL statement?

A. To sort the results of an SQL query
B. To group rows by a specific column or columns
C. To limit the number of rows returned
D. To filter the results of an SQL query

19. What are the main characteristics of the CAN bus?

A. High speed, data integrity, and bus arbitration
B. Low latency, reliability, and flexibility
C. Wide range, high bandwidth, and low power consumption
D. Simple implementation, low cost, and wide adoption

20. How do you connect the MCP2515 to a microcontroller?

A. Serial peripheral interface (SPI)
B. Inter-Integrated Circuit (I2C) bus
C. Universal asynchronous receiver-transmitter (UART)
D. Parallel data bus

21. What is the CAN bus message frame format?

A. The structure of CAN bus data packets, including message ID, data length, and error detection codes
B. The prioritization scheme for CAN bus messages
C. The method of encoding CAN bus messages into binary data
D. The technique of decoding CAN bus messages from binary data

22. What is a Zigbee coordinator?

A. A special device that initiates and manages a Zigbee network
B. A device that receives and transmits Zigbee messages
C. A device that provides routing for Zigbee messages
D. A device that manages network security

23. What is a Zigbee end device?

A. A device that communicates directly with a Zigbee coordinator
B. A device that relays Zigbee messages to other devices in the network
C. A device that provides routing for Zigbee messages
D. A special device that initiates and manages a Zigbee network

24. What is Zigbee mesh networking?

A. A type of network topology in which devices can relay messages for each other, extending the range of the network
B. A method of connecting Zigbee devices to a central hub
C. A technique for encrypting Zigbee messages
D. A way to configure Zigbee devices for different applications

25. What is a common attack vector for IoT devices?

A. Botnets
B. Denial-of-service attacks
C. Man-in-the-middle attacks
D. All of the above

26. What is the role of risk assessment in IoT security management?

A. To identify and prioritize IoT security risks
B. To implement security controls to mitigate risks
C. To monitor and assess the effectiveness of security controls
D. To respond to security incidents

27. What is the purpose of a security gateway in IoT deployments?

A. To enforce access control policies between devices and networks
B. To inspect and filter traffic between devices and networks
C. To provide centralized management of security devices
D. To monitor and log security events

28. What is the Zigbee stack?

A. A collection of software protocols that enable Zigbee devices to communicate with each other
B. A hardware component that provides Zigbee connectivity for microcontrollers
C. A network interface card that connects Zigbee devices to a network
D. A software library that implements Zigbee protocols

29. How does Kepware connect to devices in a ThingWorx environment?

A. Through direct network connection using device IP address
B. Through a data acquisition gateway
C. Through pre-defined drivers for specific protocols
D. By manual configuration of communication parameters

30. What is NOT a primary advantage of using Kepware with ThingWorx?

A. Reduced development time for industrial IoT applications
B. Increased compatibility with diverse industrial protocols
C. Enhanced data security through Kepware software
D. Improved device management within ThingWorx platform

SECTION B

Note: Attempt All Questions. **40 x 0.5 =20 Marks**

31. What are the different data types in MySQL?
    A. Integer, float, character, date, and time
    B. Boolean, string, and decimal
    C. Text, blob, and medium blob
    D. All of the above

32. What are the different data constraints in MySQL?
    A. Primary key, foreign key, and unique key
    B. Not null constraint, default constraint, and check constraint
    C. Auto-increment constraint, index constraint, and partition constraint
    D. All of the above

33. What is the purpose of an index in MySQL?
    A. To improve the performance of data retrieval by quickly locating rows in a table
    B. To enforce referential integrity between tables
    C. To prevent duplicate values from being stored in a table
    D. To optimize the performance of data modification

34. What is the purpose of a foreign key constraint?
    A. To enforce referential integrity between two tables by linking related rows
    B. To prevent orphan records from being created in the child table
    C. To ensure data accuracy and consistency across multiple tables (
    D. To improve the performance of data retrieval and manipulation

35. What is the difference between a SELECT statement and an UPDATE statement?
   A. SELECT statement is used to retrieve data from a table, while UPDATE statement is used to modify data in a table.
   B. SELECT statement uses the WHERE clause to filter data, while UPDATE statement uses the SET clause to modify data.
   C. SELECT statement returns a result set, while UPDATE statement returns an integer indicating the number of rows affected.
   D. All of the above

36. What is the difference between a prepared statement and an ordinary SQL statement?
   A. A prepared statement is precompiled and stored by the database server, improving execution efficiency, while an ordinary SQL statement is compiled and executed each time it is run.
   B. A prepared statement allows for parameter binding, reducing the risk of SQL injection attacks, while an ordinary SQL statement directly embeds parameters into the query string.
   C. A prepared statement is more flexible for handling dynamic data, while an ordinary SQL statement is simpler for basic queries.
   D. A prepared statement is primarily used for bulk data operations, while an ordinary SQL statement is used for a wider range of query

37. Which wireless communication protocol is commonly used for industrial applications due to its reliability and low power consumption?
   A. Bluetooth
   B. Zigbee
   C. Wi-Fi
   D. NFC

38. Which of the following is a key feature of edge computing in industrial connectivity?
   A. Centralized processing
   B. Low-latency data processing
   C. Dependency on cloud servers
   D. Slow response time

39. Which communication protocol is commonly used for real-time control in industrial automation?
   A. HTTP
   B. SNMP
   C. Modbus
   D. SMTP

40. Which communication standard is commonly used for wireless sensor networks in industrial applications?
   A. Bluetooth Low Energy (BLE)
   B. Infrared Data Association (IrDA)
   C. Long-Term Evolution (LTE)
   D. Wireless Fidelity (Wi-Fi)

41. In the context of industrial cybersecurity, what does the term "firewall" refer to??
   A. Physical barrier in a manufacturing facility
   B. Security software that monitors and controls network traffic
   C. Emergency shutdown system
   D. A device for measuring temperature in industrial processes

42. What is the function to create a Numpy array of ones?
   A. numpy.ones()
   B. numpy.one_array()

C. numpy.create_ones()
D. numpy.full_array()

43: Which of the following attributes of DataFrame is used to display all the values from DataFrame?
A. values
B. Values
C. val
D. Val

44. The data label associated with a particular value of Series is called ?
A. Data value
B. Index
C. Value
D. None of Above

45. How can you get the bottom n rows of a DataFrame?
A. df.tail(n)
B. df.last(n)
C. df.bottom(n)
D. df.select(n, from_end=True)

46. How can you remove duplicate rows from a DataFrame?
A. df.unique()
B. df.remove_duplicates()
C. df.filter_duplicates()
D. df.drop_duplicates()

47. Which of the following operations can be performed on a Pandas Series?
A. Filtering
B. Arithmetic operations
C. Sorting
D. All of the above

48. How can you select multiple columns in a data frame?
A. df['column1', 'column2']
B. df[['column1', 'column2']]
C. df[column1, column2]
D. df.column1, column2

49. How do you read a CSV file in pandas?
A. pandas.read_csv()
B. pandas.load_csv()
C. pandas.import_csv()
D. pandas.fetch_csv()

50. What are Pandas?
A. A programming language
B. A data analysis library in Python
C. A machine learning algorithm
D. A statistical software

51. If a dimension is given as ___ in reshaping operation, the other dimensions are automatically calculated?
A. Zero
B. One
C. Negative One
D. Infinite

52. Numpy package able to do the fast operations on arrays?
A. False
B. True
C. None of Above
D. Both a and b

53. Which of the following is the method to change the data type of an array?
A. dtype
B. datatype
C. Astype
D. astype

54. Which of the following is not an argument of the arrange method?
A. start
B. stop
C. end
D. add

55. Which of the method is used to check the dimension of an array
A. Shape
B. dim
C. shape
D. Dim

56. What is the method to find the sum of elements in a Numpy array?
A. array.sum()
B. numpy.sum()
C. array.add()
D. numpy.total()

57. What is the function to create a Numpy array?
A. numpy.array()
B. numpy.asarray()
C. numpy.create_array()
D. numpy.make_array()

58. What is the function to create a Numpy array of zeros?
A. numpy.zeros_array()
B. numpy.zeros()
C. numpy.empty_array()
D. numpy.create_zeros()

59. What is the method to find the data type of a Numpy array?
A. ndarray.item_type
B. ndarray.type
C. ndarray.data_type
D. ndarray.dtype

60. In the given code dataFrame df1 has ___rows and ____columns?

```
Import pandas as pd
Dict = {"Name":["Amit" , "Raj" , "Dev"], "RollNo":[1,2,3]}
Df1 = pd.dataFrame(dict)
```

A. 3,3
B. 2,3
C. 3,2
D. None of Above

61. Which of the following commands is used to create a series ?
A. createSeries()
B. series()
C. create_series()
D. Series()

62. A _____ is a two dimensional labeled data structure ?
A. DataFrame
B. series
C. List

D. Tuple

63. A_____ is a one-dimensional Array?
A. DataFrame
B. Series
C. Both of Above
D. None of Above

64. Which of the following is not a type of data analytics?
A. Descriptive
B. Subscripted
C. Predictive
D. prescriptive

65. Which of the following commands is used to install pandas?
A. pip install pandas
B. install pandas
C. pip pandas
D. None of Above

66. What is the method to fill missing values in a pandas DataFrame or Series?
A. update()
B. replace()
C. fillna()
D. impute()

67. What is the method to find unique values in a pandas DataFrame or Series?
A. unique()
B. nunique()
C. count_unique()
D. get_unique()

68. How can you get the top n rows of a DataFrame?
   A. df.top(n)
   B. df.first(n)
   C. df.head(n)
   D. df.select(n)

69. How can you concatenate two or more DataFrames?
   A. df.concat([df1, df2])
   B. df.append([df1, df2])
   C. df.merge([df1, df2])
   D. df.combine([df1, df2])

70. Which layer of the OSI model does CAN primarily operate in?
   A. Physical Layer
   B. Data Link Layer
   C. Network Layer
   D. Transport Layer

Course Name: Operating Systems

Course Outcome

CO1- Understand the classification of operating system environment.
CO2- Understand the basic of process management.
CO3- Apply the concept of CPU process scheduling for the given scenarios.
CO4- Illustrate the process synchronization and concurrency process in operating system.
CO5- Analyze the occurrence of deadlock in operating system.
CO6- Describe and analyze the memory management and its allocation policies.
CO7- Understand the concepts of disk scheduling

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End Term Examination, Odd Semester 2023-24
B.Tech. (CSE), AIML, CCV, DA, CSF, IIOT & EC (Minor in CS), Year-II, Semester-III
OPERATING SYSTEMS (BCSC 0004)

Time: 3 Hours **Maximum Marks: 50**

Note: Attempt all Sections

Section – A

*Attempt **All** Questions* 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Describe the differences between symmetric and asymmetric multiprocessing. What are three advantages and one disadvantage of multiprocessor systems?	2+2	1	R	F
2	How operating system manages critical section problem by using semaphore? Differentiate between binary and counting semaphore? Write down the Peterson solution for two process synchronization.	1+1+2	4	U	C
3	An operating system uses the banker`s algorithm for deadlock avoidance when managing the allocation of three resource types X, Y and Z to three processes P0, P1 and P2. The table given below presents the current system state. Here, the Allocation matrix shows the current number of resources of each type allocated to each process and the Max matrix shows the maximum number of resources of each type required by each process during its execution.	2+2	5	A	P

		Allocation			Max					
		X	Y	Z	X	Y	Z			
	P0	0	0	1	8	4	3			
	P1	3	2	0	6	2	0			
	P2	2	1	1	3	3	3			
	There are 3 units of type X, 2 units of type Y and 2 units of type Z still available. The system is currently in safe state. Consider the following independent requests for additional resources in the current state- **REQUEST1**: P0 requests 0 units of X, 0 units of Y and 2 units of Z **REQUEST2**: P1 requests 2 units of X, 0 units of Y and 0 units of Z Find out whether REQUEST1 and REQUEST2 can be permitted or not?									
4	Consider the 3 processes, P1, P2 and P3 shown in the table: Process — Arrival time — Time Required P1 — 0 — 5 P2 — 1 — 7 P3 — 3 — 4 Find the completion order and calculate average waiting time of the 3 processes under the policies FCFS and RRS (round robin scheduling with CPU quantum of 2 time units).							2+2	3	A	P
5	Why translation lookaside buffers are necessary? A paging scheme uses a Translation Lookaside buffer (TLB). The effective memory access takes 260 ns and a main memory access takes 100 ns. What is the TLB access time (in ns) if the TLB hit ratio is 80% and there is no page fault?							1+3	6	U	C

Section – B

*Attempt **All** Questions* 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Draw the resource allocation graph and determine deadlock or deadlock free state. P = {P1, P2, P3} R = {R1, R2, R3, R4} E = { P1 →R1, P2→R3, R1→P2, R2→P2, R2→P1, R3→P3, P3→R2} There are ONE, TWO, ONE and THREE instances of R1, R2, R3 and R4 type resources	2+1	5	An	C
2	Consider six memory partitions of size 200 KB, 400 KB, 600 KB, 500 KB, 300 KB and 250 KB. These partitions need to be allocated to four processes of sizes 357 KB, 210 KB, 468 KB and 491 KB in that order. Perform the allocation of processes using- 1. First Fit Algorithm 2. Best Fit Algorithm 3. Worst Fit Algorithm	1+1+1	6	A	P
3	The Logical memory and physical memory space is 32k and 512 words respectively. If Page size is of 64 words, find the logical and physical address. How many pages logical and physical memory can accommodate? Write number of entries in the page table also.	1+1+1	6	E	P
4	A process has been allocated 3 page frames. Assume that none of the pages of the process are available in the memory initially. The process makes the following sequence of page references (reference string): 1, 2, 1, 3, 7, 4, 5, 6, 3, 1 If optimal page replacement policy is used, how many page faults occur for the above reference string?	3	6	E	C
5	A disk with 1000 cylinders, 0 to 999, compute the number of track a disk arm must move to satisfy all the request in the disk queue. Assume the last request served was at track 345 and head is moving towards 0. The queue in the FIFO order is	1.5+1.5	7	A	P

	123, 874, 692, 475, 105, 376 Perform the computation for the following Disk Scheduling Algorithm. a) SSTF b) SCAN				

Section – C

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Discuss Linked and Indexed File allocation methods with its advantages and disadvantages. Which allocation scheme will minimize the amount of space required in the directory structure and why?	3+2	7	U	F
2	Consider the disk system with 100 cylinders. The request to access the cylinders occur in the following sequence. 4, 37, 10,7,19,73,2,15,6,20 Assuming the head is currently at cylinder 50 what is the time taken to satisfy all requests if it takes 1ms to move from one cylinder to adjacent one and Scan and C- Look algorithms are used.	2.5+2.5	7	A	P
3	Assume **"Department of CEA, GLA University"** implemented the new page replacement algorithm in virtual memory and given its name as 'CEAR'. Consider the working strategy of CEAR as following- Each page in memory maintains a count which is incremented if the page is referred and no page fault occurs. If a page fault occurs, the physical page with zero count or smallest count is replaced by new page and if more than one page with zero count or smallest count then it uses FIFO strategy to replace the page. Find the number of page faults using CEAR algorithm for the following reference string (assume three physical frames are available which are initially free) and the Reference String: "A B C D A B E A B C D E B A D"	5	6	C	M

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End Term Examination, Session 2023-24

B.Tech (CSE, AIML,CCV, DA, CSF, IIOT), 2nd Year, III - Semester

BCSC 1002 – Object Oriented Programming

Time: 3 Hours **Maximum Marks: 80**

Instruction for students:
1. Read questions Carefully.
2. Answer the Question in the Question paper
3. In the fill the blanks, ';' means end of the statement.
4. Write only the dedicated space for each.
5. Attempt all questions.

Section – A (30 Marks)

1. Choose the Correct Option for the following questions. **(10 marks)**

A. What is the output of the following?

```
String b = "12";
b += "3";
b.reverse();
System.out.println(b.toString());
```

a. 12
b. 123
c. 321
d. The code does not compile

Answer:_____

B. What is the output of the following?

```
String line = new String("-");
String anotherLine = line.concat("-");
System.out.print(line == anotherLine);
System.out.print(" ");
System.out.print(line.length());
```

a. false 1
b. false 2
c. true 1
d. true 2
e. Does not compile

Rough work Student's Signature:

```
    { super(100);
        System.out.println("child-no-arg-
const");
    }
    public static void main(String []args)
    {
        Child c=new Child();
        System.out.println(c.x);
    }
}
```

Answers:

A - _____

B - _____

C - _____

D - _____

E - _____

Rough Work

Rough work Student's Signature:

Course Name: **BCSE 1502 & Introduction to Virtualization and Cloud Computing**

Course Outcome

CO1: Understand the basics of virtualization technology and hypervisors.
CO2: Analyze the trade-offs between deploying applications in the cloud and over the local infrastructure.
CO3: Recognize real-world problem using cloud computing through group collaboration.
CO4: Understand on-demand utility computing phenomenon of cloud computing.
CO5: Understand the issues involved in cloud computing.

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End Term Examination, Odd Semester 2023-24
B.Tech (CCV),2nd Year, 3rd Semester
BCSE 1502 & Introduction to Virtualization and Cloud Computing

Time: 3 Hours **Maximum Marks: 50**

Instruction for students: Attempt all the questions.

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Compare traditional IT infrastructure with virtualized infrastructure in cloud computing. Highlight four key differences between these approaches and briefly explain how each impacts scalability and resource management.	4	1	R	F
2	How does the adoption of virtualization impact both the cost and manageability of IT infrastructure within an organization? Provide an analysis detailing the effects of virtualization on cost savings and improved manageability. Illustrate at least four key aspects that highlight the financial benefits and enhanced operational control achieved through the implementation of virtualization technology.	4	2	U	P
3	A Corporation, a growing enterprise, is facing challenges with their existing IT infrastructure. They are experiencing high operational costs, inefficiencies in resource utilization, and difficulties in managing their expanding network. As an IT consultant, explain the scenario-based need for implementing virtualization within this Corporation. Provide four key reasons outlining why virtualization would address their challenges, and briefly explain the significance of each reason.	4	3	R	C
4	Compare and contrast Cloud Computing and Grid Computing, elucidating their individual characteristics,	4	4	U	F

	functionalities, and their interrelation in modern computing paradigms. Provide an in-depth analysis covering their similarities, differences, and how they complement each other within contemporary IT architectures.				
5	DEF Enterprises, a multinational corporation, is planning to migrate its IT infrastructure to the cloud. As a cloud computing consultant, explain the scenario-based considerations for selecting an appropriate deployment model for DEF Enterprises. Provide four key factors that DEF Enterprises should evaluate when choosing a deployment model, and briefly explain the significance of each factor. Or GHI Tech Solutions, a startup specializing in software development, is exploring options to host its applications in the cloud. As a cloud computing advisor, outline scenario-based considerations for GHI Tech Solutions to choose an appropriate cloud service delivery model. Provide four key factors that GHI Tech Solutions should evaluate when selecting a delivery model, and briefly explain the significance of each factor.	4	3	An	C

Section – B

*Attempt **All** Questions* 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Outline three specific theoretical ways in which cloud computing can address their resource limitations, improve operational efficiency, and stimulate innovation.	3	2	A	P
7	VM Finance, a financial services firm, deals with occasional intensive computational workloads, particularly during quarterly financial reporting periods. They aim to minimize costs while ensuring their on-premises infrastructure can handle these periodic spikes. Explain how cloud bursting can support VM Finance in addressing these peak computational demands efficiently. Outline the concept of cloud bursting and how it enables VM Finance to seamlessly access additional compute resources from the cloud when needed, optimizing performance without incurring unnecessary infrastructure costs.	3	4	An	C
8	Discuss the ways in which virtualization technologies can enhance disaster recovery capabilities and bolster system reliability.	3	4	U	F

9	In what ways does geographical location play a crucial role in cloud deployment decisions? Provide three intricate considerations related to geographic factors that organizations often overlook but significantly influence the choice of cloud deployment strategy. Elaborate on the implications of these considerations on cloud performance and compliance.	3	5	A	C
10	Provide recommendations and security measures that a cloud platform should consider to mitigate risks and ensure the protection of their sensitive financial data and transactions.	3	4	An	P

Section – C

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	Explain the key components and critical steps involved in creating a comprehensive cloud transformation roadmap for an enterprise. Outline a detailed plan encompassing five essential stages that organizations should consider when developing and implementing a cloud transformation strategy. Elaborate on the significance of each stage in ensuring a successful and smooth transition to the cloud.	5	4	R	F
12	Describe the cultural and economic factors that contributed to the widespread adoption and acceptance of cloud computing within businesses and industries. Analyze the shift in attitudes toward cloud technologies and the driving forces behind their mainstream acceptance in modern IT landscapes.	5	5	An	F
13	You're an IT consultant working with a multinational conglomerate that operates across diverse sectors, including finance, healthcare, retail, and manufacturing. The company is considering the implementation of virtualization technologies and evaluating the migration of its workloads to the cloud. As part of your role, you're tasked with advising the company on the strategic aspects of virtualization implementation and determining the key factors influencing the decision-making process for migrating workloads to the cloud across these varied sectors.	5	4	E	C

Course Name: Introduction to Business Analytics

Course Outcome

CO1: Understand the use of reference architecture in business analytics.
CO2: Explain the decision support science and data warehouse.
CO3: Understand the concepts of Business Intelligence.
CO4: Explain the KDD in data mining.
CO5: Understand the process of analysis using Dashboards and Reports.
CO6: Explain Big Data, Filtering BigData and Adoption Architecture.

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End Term Examination, Odd Semester 2023-24
B.Tech (DA), Year - II, Semester - III
BCSE0551: Introduction to Business Analytics

Time: 3 Hours

Maximum Marks: 50

Instruction for students: Attempt all questions

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Discuss various classification of analytics with examples.	4	4	U	F
2	Shyamli is working for Ministry of finance at New Delhi. She was assigned with a task to collect information about transactions above ₹49000. She is suppose to collect data from Share Markets, Banks, Fixed Deposits, Property Deals, Expenditure on Foreign Trips, Online Shopping and others. All these various sources had varied formats, column, their types. Aggregating whole of this data to a single file for each PAN number is she want before loading to data warehouse. Explain the process she needs to perform before loading the data to data warehouse.	4	1	A	P
3	What is data cube? Explain the following operations with examples: a. Drill down b. Roll-up c. Dice d. Slice	4	2	U	F
4	You work in New York based share trading firm. Your client now excited to trade in NFTs and want you to build a marketplace for it. You are supposed to design the multi-tier architecture for data	4	5	U	F

	warehouse of this client. Draw and explain the multi-tier architecture of data warehouse.				
5	Consider yourself working for firm in Delhi. Explain the following to your boss: a. Data Mining b. CRISP c. Machine Learning d. Reinforcement Learning	4	3	R	F

Section – B

*Attempt **All** Questions* 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	You work in analytics firm and had their business data of sales for last quarter in `sales.csv`. Perform the following in python. a. Write code to read this file to pandas Dataframe. b. To check sales data and since you can't have look at whole data, write code to display 50 rows *randomly*. c. Since `sales.csv` has only 300 rows, and you want to stretch this data to 4 quarters (i.e. 4 times), write code to display 1200 rows *randomly*. d. Display top 50 rows. e. Display last 50 rows.	3	3	C	M
7	Discuss ROLAP, MOLAP and HOLAP. Differentiate between OLTP and OLAP. What are the features of data warehouse?	3	6	U	F
8	How missing values is represented? How you can deal with missing values? Explain all steps for treating missing values?	3	2	R	F
9	Answer the following: a. Define Big Data b. What are its Vs. Explain with example. c. List any three tools used to handle big data	3	3	U	C
10	As an analyst you need to deal with various issues of data collected of various ***business operations***. Explain the following:	3	6	E	P

	a. How to tell python that this variable is categorical? Why it is necessary to tell python about this? b. How to create ordinal categorical in python? c. Consider you have a list or pandas serious of 50 numbers between 0 to 1. How will your convert this data to 4 categories with precision of 2 decimal places.				

Section – C

*Attempt **All** Questions*

5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	Consider you have a file kpi.csv. write code for following: a. Import this file as pandas DataFrame. b. On analysis, you found a blank space ' ' was used in various places for missing value. Tell python that space is missing value and reimport the file. c. In column cashinflow, at many places 0 is mentioned, which is semantically wrong. Convert these 0 to Nan. d. In column sales_agent, few places has null values, perform listwise deletion on sales_agent. e. In column avg_sales, impute null vales by mean.	5	6	E	M
12	Consider you have one file for product_information and another file for product_sales. a. Import both files in pandas DataFrame both are in csv format. b. Display first 5 rows of both DataFrame. c. How will you merge both DataFrame so you get whole data in one single DataFrame. You need to perform merging on product_id which is present in both DataFrame. d. How will you merge both DataFrame so resultant DataFrame will also have information of products not yet sold. e. Clerk had performed some mistakes while entering data and randomly put product_id in	5	2	C	M

	sales file. How will you perform merging so that resultant DataFrame will be having information product sales which actually have no existence in product information.				
13	As a digital marketing firm employee, your client came to you, with the problem of email bouncing, low views. Explain him with the types of *dashboards* available. Also, conclude which of these *dashboards* will suit his requirement the most.	5	5	A	P

Course Name: IT Application and Data Security

Course Outcome

CO1-Understand software development methodology and application security
CO2-: Explain input validation strategies; protect sensitive data, effective authentication and authorization strategies.
CO3- Understand various attacks like DoS, buffer overflow, web specific and database specific.
CO4- Implement security as a culture to show mistakes that make applications vulnerable to attacks.
CO5- Demonstrate the security enhancements of web based applications.
CO6- Understand the basics of data security threats.
CO7- Apply the data security threat techniques and its counter measures

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End Term Examination, Odd Semester 2023-24

Program: B. TECH(CSF), Year: II, Semester: III

Subject Code & Name- IT Application and Data Security (BCSE 0601)

Time: 3 Hours **Maximum Marks: 50**

Section – A

*Attempt **All** Questions* 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	You are a project manager leading a team in the software development life cycle (SDLC) for implementing a cyber-security solution in an organization. The goal is to enhance the security posture of the organization's web applications, databases, and network infrastructure. The organization has experienced recent security incidents, and the leadership is keen on addressing vulnerabilities and ensuring a robust cyber-security posture. Suggest the solution for same.	4	1	R	F
	Why input validation is necessary in web based application? Suppose you are a software developer working on a social media platform. The platform allows users to sign in using their email and password. However, you want to enhance the authentication process by implementing two-factor authentication. How can two-factor authentication improve the security of the platform?	4	2	U	C
3	How would you secure session data to prevent unauthorized access? Explain any modern best practices regarding session management security? What is cookie hijacking and discuss the various Prevention methods for cookie hijacking.	4	3	An	F

4	Differentiate between Auditing & Logging. Use an affine cipher to encrypt the message "life is full of surprises" with the key pair (7, 2) in modulus 26. Use encoding of 00 to 25 for letters A to Z respectively.	4	4	U	F
5	Alice and Bob wish to share private message, where each of them of two separate keys generated. What kind of strategy would you suggest to ensure confidentiality, key management, authentication and integrity for the conversation between Alice and Bob? Explain the strategy and also highlight the design issues related to the strategy proposed. Or Malware, known as Stuxnet is a shorthand term that encompasses all types of malicious software. This includes viruses, worms, Trojan horses, spyware, and all other types of software that get put onto your computer without you knowing it. Malware may exploit weakness to initially infiltrate a system or to gain additional privileges on an already-compromised machine. The weaknesses may be exploiting automatically by malware authors creations or manually by attempt will be generically called attackers. Weaknesses fall into two broad categories, based on where the weakness lies. Technical weakness involves tricking people. The malware, hidden in shortcuts to executable programs (files with extension .ink) was executed automatically when the content of an infected USB drive was displayed. Employing this same technique, the worms were capable of installing other malware. Data gathered later by other experts indicates that the worms was actually looking for some specific programmable Logic Controllers(PLC) devices used in a specific	4	4	A	C

	industrial plants. A fact that points to the possibility that malware was part of the sites infected with sabotage. The significance that such a sophisticated treat represents to the industrial recourses in Europe and others parts of the world cannot be underestimated. Q.1 Describe some of the weaknesses exploited by malware. Q.2 Discuses the steps of malware threat technique in correct order.				

Section – B

*Attempt **All** Questions* 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Why data security is play a crucial role for any organization? Using an appropriate diagram, describe and discuss the data security lifecycle.	3	6	R	F
7	The government security paradigm was "closed and locked," whereas the academic security model was "wide open." Provide a relevant example to support this claim.	3	6	U	C
8	Differentiate between Birthday attack and cryptanalysis attack. What are the steps involved in cross-site scripting forgery.	3	7	R	F
9	Interception can be countered with Secure Socket Layer (SSL) and advanced private-key protection. These ensure that the data traversing between the client and the server cannot be read in clear text, if intercepted. So discuss the working of following protocol: a. SSL Record protocol b. SSL Alert protocol c. SSL handshake protocol	3	7	U	C
10	You work as a cyber-security analyst for a medium-sized company. One morning, the company's IT department notices a significant slowdown in network performance. Users are	3	5	A	C

	reporting difficulties accessing critical applications, and there's an unusual increase in network traffic. As you investigate, you discover that the company's servers are under a Distributed Denial of Service (DDoS) attack. Upon further analysis, you identify that the DDoS attack is being orchestrated by a botnet. The botnet is using a variety of attack vectors, including a mix of HTTP GET floods and DNS amplification attacks. The attack is causing disruptions to the company's online services, impacting both internal and external users. a. What steps would you take to immediately mitigate the impact of the DDoS attack on your company's servers? b. What tools or technologies could you employ to detect and mitigate the different types of DDoS attacks within the network? c. What proactive measures could the company implement to prevent future botnet-driven DDoS attacks?				

Section – C

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	Network security plays a significant role to protect your private network from various types of threats. Discuss and explain following counter measures to protect you private network: a. Intrusion Detection Systems(IDS) b. Demilitarized Zone(DMZ) c. Security Logs/Access Logs d. Intrusion Prevention System(IPS) e. Network Monitors & Honeypot	5	7	A	C
12	Why is social engineering a significant threat to cyber-security? Explain the following types of social engineering attacks with suitable example and preventive measures: a. Baiting b. Phishing	5	6	U	F

	c. Pre-texting d. Spear phishing e. Tailgating				
13	TMC Corporation is a multinational company specializing in technology solutions and services. With a vast network of clients and a significant online presence, the company handles sensitive data, including customer information, financial records, and proprietary technology. Despite implementing various security measures, TMC Corporation recently experienced a data security breach that compromised a substantial amount of confidential information. On a Monday morning, the IT department at TMC Corporation discovered unusual activity on their servers. Investigation revealed that an unauthorized party had gained access to the company's database, leading to the compromise of sensitive customer data, employee records, and intellectual property. The company promptly initiated an incident response plan, involving IT security experts, legal counsel, and communication specialists. 1. How did the security breach occur? 2. What steps did TMC Corporation take immediately after discovering the breach? 3. How did TMC Corporation communicate the breach to affected parties and the public? 4. What measures will TMC Corporation implement to prevent future breaches? 5. What data was compromised?	5	7	U	C

Course Name: Statistics and Complex Analysis

Course Outcome:

CO 1: Compute moments and use them to visualize symmetry and peakedness of the data

CO 2: Fit a straight line on the given data by the method of Least squares

CO 3: Measure the extent of correlation between two variables

CO 4: Predict the value of a variable by finding regression lines

CO 5: Apply discrete probability distributions to the real-life data problems

CO 6: Apply Chi-square test to test a statistical hypothesis in real-life data analysis

CO 7: Know the concepts of analytic and harmonic functions and their applications

CO 8: Solve problems involving complex integration

CO 9: Find radius and region of convergence of a power series

CO 10: Calculate residue and apply Cauchy's residue theorem in solving real integrals

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End Term Examination, Odd Semester 2023-24

B. Tech. (CS, CS – Spec., EC, EC- Spec., EE, CE), II-Year, III-Semester

BMAS 0106: Statistics and Complex Analysis

Time: 3 Hours **Maximum Marks: 50**

Instruction for students:

1. Attempt all questions.
2. Solution must be brief, complete and to-the-point.

Section – A

Attempt All Questions. 4 X 5 = 20 Marks

<table>
<tr><th>S.No.</th><th>Detail of Question</th><th>Marks</th><th>CO</th><th>BL</th><th>KL</th></tr>
<tr><td>1</td><td>What do you mean by correlation? Give some real life examples of correlation between two variables. Calculate Karl Pearson's coefficient of correlation between x and y from the following data:
<table><tr><td>x</td><td>1</td><td>3</td><td>5</td><td>7</td><td>8</td><td>10</td></tr><tr><td>y</td><td>8</td><td>12</td><td>15</td><td>17</td><td>18</td><td>20</td></tr></table></td><td>4</td><td>3</td><td>U</td><td>F</td></tr>
<tr><td>2</td><td>What are regression lines? Write their equations. Find the regression line of y on x from the following data:
<table><tr><td>x</td><td>1</td><td>3</td><td>5</td><td>7</td><td>9</td></tr><tr><td>y</td><td>15</td><td>18</td><td>21</td><td>23</td><td>22</td></tr></table></td><td>4</td><td>4</td><td>U</td><td>F</td></tr>
</table>

3	In a sample analysis of examination results of 500 students, it is found that 220 students failed, 170 secured a third class, 90 secured a second class and the rest, a first class. Do these figures support the general belief that above categories are in the ratio 4: 3: 2: 1 respectively? [Given that the tabular value of χ^2 for degree of freedom 3 at 5 % level of significance is 7.81.]	4	6	A	C
4	Define 'Analytic Functions'. Show that the function $f(z)=e^{z}$ is analytic everywhere in finite complex plane.	4	7	U	C
5	Show that for the function, $f(z)=\begin{cases}\dfrac{x^2y^5(x+iy)}{x^4+y^{10}}, & z\neq 0\\ 0, & z=0\end{cases}$ the Cauchy-Riemann equations are satisfied at the origin but derivative of $f(z)$ does not exist at origin.	4	7.	An	P

Section – B

*Attempt **All** Questions.* 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Determine an analytic function $f(z)$ in terms of z whose real part is $e^{-x}(x\sin y - y\cos y)$.	3	7	E	C
7	Prove that the functions, $u=\frac{1}{2}\log(x^2+y^2)$ and $v=tan^{-1}\frac{y}{x}$ are harmonic conjugates of each other.	3	7	U	C

8	Evaluate: $\int_0^{2+i} (\bar{z})^2 \, dz$, along the real axis from $z = 0$ to $z = 2$ and then along the line parallel to y-axis from $z = 2$ to $z = 2 + i$.	3	8	A	P
9	Find the radius of convergence of power series $\sum_{n=0}^{\infty} \left(\frac{z^n}{2^n + 1} \right)$	3	9	U	C
10	Expand the function $f(z) = \frac{1}{(z+1)(z+3)}$ in the following regions: (a) $1 < \lvert z+1 \rvert < 2$ and (b) $1 < \lvert z \rvert < 3$.	3	10	An	P

Section – C

*Attempt **All** Questions.* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	Determine poles and residue at each pole of the function $f(z) = \frac{z^2}{(z-1)^2(z+2)}$. Hence evaluate $\int_C \frac{z^2}{(z-1)^2(z+2)} \, dz$, where $C \equiv \lvert z \rvert = 3$.	5	10	An	P
12	Discuss the various types of singularities in a function of complex variable and give suitable examples.	5	7	U	C

13	**Case study:** Cauchy Integral theorem states that "if a function F (z) is continuous at each point within and on a simple closed curve C then integral of that function along the curve C remains zero". However, Goursat later proved that the assumption of continuity of derivative of f (z) is unnecessary and Cauchy's theorem holds if f (z) is analytic within and on the curve C. It was made applicable to multi-connected regions by converting multi-connected regions to simply connected regions. Subsequently, Cauchy used the concept of residues to propose the Cauchy residue theorem which states that: $$\oint_c f(z)\,dz = 2\pi i\,(R_1 + R_2 + R_3 + \cdots)$$ where R_1, R_2, R_3,... are the residues of f (z) at the poles lying within the curve C. **Based on the above case study, evaluate the following complex integral,** $$I = \oint_c \frac{dz}{(z^2+1)(z^2-4)}$$ where, (a) $C \equiv \lvert z \rvert = 0.5$ (b) $C \equiv \lvert z \rvert = 2$ (c) $C \equiv \lvert z - i \rvert = 1$ (d) $C \equiv \lvert z + i \rvert = 1$ (e) $C \equiv \lvert z \rvert = 5$	5	8,10	E	P

Course Name: English For Professional Purposes I

Course Outcome

CO1- create structurally effective, grammatically accurate and contextually appropriate written and spoken pieces for communication both independently and collaboratively.

CO2-display competence in all aspects of language learning not limited to listening, speaking, reading and writing.

CO3-develop independent perspectives and points of view and express them in a cohesive and meaningful way.

CO4-produce a repertoire of register-specific and academic vocabulary appropriate in both oral and written communication.

CO5-use language as an effective tool of communication for various contexts in both academic and professional setting.

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End Term Examination, Odd Semester 2023-24

B.Tech II Year, III Semester (All Branches) & MCA I Year, I Semester

BELH 1003 English for Professional Purposes I

Time: 3 Hours **Maximum Marks: 50**

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Mark	CO	BL	KL
1	i-Identify the grammatical function of the word fast in each of the following sentences 1- By the time I got home, the children were already fast asleep. A. Noun B. Verb C. Adverb D. Preposition 2- Some prisoners began a fast to protest against the appalling conditions A. Verb	4	1	A	F

	B. Adjective C. Noun D. Interjection i-Kelly has great confidence _____ her children's ability. Choose the correct preposition. A. on B. at C. for D. in iii. Unfortunately, many online courses do not provide students _____ the basic skills they need. A. with B. for C. on D. in iv. They hope to can arrive before the start of the conference. Identify the unnecessary word in the sentence and its part of speech. A. can- linking verb B. to- infinitive C. the (second) – Definite article D. can - Auxiliary Verb v. J: I don't have any money. K: I don't have no money. L: I don't have money. The grammatically correct sentences are: A. J and K B. K and L C. J and L D. J, K and L vi. Leap year falls in _____ forth year. A. Each B. Every C. One				

	D. Any vii. He showed _____ concern for his nephew. A. the little B. a few C. little D. a little viii. Whether I shall turn out to be the hero of my own life, or whether that station will be held by anyone else, _____ pages must show. A. these B. this C. none D. third ix. _____truth must be spoken. A. a B. the C. no article D. Both A and B. x.In which year were both these articles circulated to all the students? Identify the determiners. A. to, all, these, both. B. both, the, all, in. C. to, year, all, the. D. these, the, both, all xi. Linear communication is _____ one-way process conducted in ____ straight line towards ____ main point. Fill in the blanks with correct determiners. A. a, the, the B. the, a, a C. a, the, a D. a, a, the xii. Most men and women will grow up to love _____				

servitude and will never dream of revolution.

Fill in the blank with the correct determiner.

A. our
B. there
C. their
D. Many

xiii-The manager said to me, "Complete it by tomorrow."

A. The manager requested me to complete it by tomorrow.
B. The manager instructed me to complete it by the next day.
C. I was ordered by the manager to complete the work by the next day.
D. The manager ordered me to complete the work by the next day.

xiv- J) Today it was first time we play such games.
k) Bacteria can kill you into a few days.
L) I am leaving a call at 10:00 AM.

The grammatically correct sentence is:

A. J and K
B. K and L
C. J and L
D. None of the above

xv- J) I will buy some mangoes.
K) I will not buy some mangoes.
L) A little knowledge is a dangerous thing.
M) A darkest cloud has a silver lining.

Which of the following sentences have the correct use of determiners?

A. J and L only
B. K, L and M are correct
C. Only J is correct
D. Only M is correct

xvi- He gave a talk _____ the causes of Dengue.

A. on

	B. for C. at D. of				
2	i He is an *absolutely honest* politician. A. Noun Phrase B. Prepositional Phrase C. Adjective Phrase D. Adverb Phrase ii.*In the realm of quantum mechanics*, researchers delve into the mysteries of subatomic particles. A. Prepositional Phrase B. Verb Phrase C. Adverb Phrase D. Independent Clause iii. The arrival *of the train* has been delayed. A. Adverb Phrase B. Prepositional Phrase C. Noun Phrase D. Verb Phrase iv. *All my sons* are good football players. A. Noun Phrase B. Adjective Phrase C. Adverb Phrase D. Verb Phrase v. It *has been raining* continuously for the last seven days. A. Verb Phrase B. Adverb Phrase C. Adjective Phrase D. Noun Phrase vi. The men in the red scarves dance *quite merrily* in the sun. A. Verb Phrase B. Noun Phrase C. Adverb Phrase D. Adjective Phrase vii.The people look solemn, yet they go to work every day. A. Independent Clause followed by a Dependent Clause. B. Dependent Clause followed by an Independent Clause. C. 2 Independent Clauses	4	1	U	F

	D. 2 Dependent Clauses viii. *Although the train arrived on time*, we still missed it. A. Dependent Clause B. Independent Clause C. Prepositional Phrase D. Verb Phrase ix. Jane's reputation as a/an _____ grew as she consistently prioritized her own needs over those of her colleagues. A. egoist B. versatile C. flamboyant D. selfish x. Before responding, he took a moment to _____, the impact of his words on the conversation. A. contemplate B. introspect C. adapt D. accustom xi. When choosing mutual funds, individuals often prefer those with a ____ history of reliable returns and performance. A. bankrupt B. bankable C. prominent D. amateur xii.The old computer system became _____ after the company upgraded to the latest technology. A. famous B. redundant C. obsolete D. obliterate xiii. The Indian Army _____ INS Viraat in the year 2017. A. decommissioned B. simulate C. reinstitute D. rehabilitate xiv. The math _____ effortlessly solved complex problems that left even seasoned mathematicians impressed. A. novice B. servile				

	C. gifted D. prodigy xv. The jury decided on the capital punishment _____ A. impromptu B. solely C. unanimously D. servile xvi. The _____ of the situation was that the more she tried to control time, the more it seemed to slip away from her grasp. A. paradox B. agenda C. veteran D. fanciful				
3	i. Rewrite the following *para* to make it coherent. "Limited investment in the housing sector makes it practically impossible to allocate sufficient resources for urban dwellers' housing needs. A high rate of urban population growth has increased the country's needs for housing. A small group of city officials has laid out a new plan to combat the crisis. A solution to the housing-shortage problem is a vital policy issue here. The housing problem has grown in the last twenty years." ii. Fill in the blanks with the most suitable transitions from the list. Use each transition only once: in spite of, thanks to, although, similar to, like, also, in addition to, whereas, compared to, A._____ we have scientific evidence that women should not smoke when they are pregnant, many future mothers ignore this danger. These mothers endanger the physical health of their babies. B-_____, they could be contributing to even more problems in the future. C- _____ the known problems, recent studies have linked mothers' smoking during their pregnancy with sons who go on to become criminals. Researchers in Finland followed over 5000 males and their mothers from the time the mothers were six months pregnant until the young men	2+2	2	A	C

	were 28 years old. D- _____ sons whose mothers did not smoke during pregnancy, those young men whose birth mothers did smoke were more than twice as likely to commit criminal acts. The researchers theorized that smoking during pregnancy may alter the chemical balance in the baby's brain.				
4	Read the following passages and answer the questions that follow: About once in every seven years, the ocean surface off the coast of Peru warms up. This cuts the normal enriching of nutrient-rich cold water. Plankton production is drastically reduced. This phenomenon, known as El Nino (the child) because it starts during the Christmas season, usually lasts for up to a year. Occasionally, it goes on for long. Three times this century, it has persisted into a third year. The latest of these prolonged episodes have been blamed for much more than its effect on the Peruvian fishery. It certainly seems to have played a part in droughts in normally humid Indonesia as well as those that brought catastrophic fires to the outskirts of Sydney. It has been blamed for storms and landslides in coastal regions of Peru and Equator and is associated with drought in north-eastern Brazil. Its influence may stretch as far as western Europe where the recent winter brought heavy rainfall and flooding. El Nino is not the only ocean phenomenon to affect the weather. To understand how this might be done, it is useful to think about how the weather is forecast and about what makes it predictable. Weather forecasting uses two types of techniques, both of which rely on observations of what is going on at and shortly before the time of the forecast. To supply this information, national authorities operate extensive land, sea and air based observation networks. These are coordinated under the United Nations World Meteorological Organisation. There is an effective and almost instantaneous worldwide exchange of the information gathered. The first, classical forecasting method compares the set of observations with experience and bases the forecast on what has happened in the past from a similar starting point. The second, only possible since the advent of fast computers, applies the laws of physics through a numerical model of the atmosphere to	4	2	U	F

predict how the weather will develop from the observed starting point. This method is the basis of modern weather forecasting. But neither method will produce useful predictions for more than a few days ahead. This is because the behaviour of the atmosphere is governed in the medium term by energy exchanges with the land and ocean. The latter is much more important as the sea surface stores energy which it exchanges with the atmosphere in a surface layer 100 m or so thick. The energy storage capacity of the layer is many times greater than that of the whole atmosphere. This had prompted the suggestion that longer term forecasting might be practicable if a coupled computer model of the atmosphere and ocean could be constructed.

A great deal of research has been put into this with some success but progress has been limited by two factors. The first is that coupled ocean-atmosphere models require enormously greater computing capacity than is provided even by the super computers used in weather forecasting. The second is that not enough is known about the state of the ocean at any given time - there is no global observation network as there is for the atmosphere–or about the processes that govern the interactions. When these limitations have been overcome, it seems likely that coupled models will permit prediction of such climatological factors as frequency and intensity of rainfall for seasons and perhaps years ahead.

In the meantime, climatologists have begun to be able to predict the onset and consequences of phenomena such as El Nino using techniques like those used in the classical weather forecasting method. These examine the condition of the ocean at a given instant and, by comparison with past experience, attempt to predict in a statistical way how the ocean or the atmosphere—and hence the climate—are likely to behave for the next few months. As with the weather forecasts, this depends critically on the existence of a bank of past experience, in this case, particularly of sea-surface temperatures. A key contribution to this, recently completed by the UK Meteorological Office, has been the analysis of many millions of sea-surface temperatures. This has led to the publication of the first globally complete monthly fields of sea-surface temperature from 1871 to the present day. This

type of information may be used to predict events such as El Nino.
Various scientific groups have tried to do so with varying degrees of success. In the meantime, Meteorological Office scientists have also compared the temperature fields statistically with climatological factors. They have shown that, particularly in the tropics, there are significant correlations between sea-surface temperature anomalies and climate statistics. This does not necessarily mean that one causes the other, though some degree of direct linkage seems likely. But it does open up the possibility of predicting short-term climate fluctuations.
The UK Meteorological Office has approached this by a rigorous comparison between rainfall statistics in the Nordeste area of Brazil, whose crops can be seriously affected by drought, with contemporaneous sea-surface temperatures world-wide. This has revealed significant links, verified over the period 1901-85, between rainfall and sea-surface temperatures in the north and south tropical Atlantic and the western tropical Pacific, the area most strongly affected by El Nino.

1. Which of the following is NOT TRUE about 'El Nino'?
(1) It is the most important ocean phenomenon to affect a region's weather pattern.
(2) There seems to be a statistical link between sea-surface temperatures and the occurrence of El Nino.
(3) The consequences of El Nino can vary with the geographical position of the affected area.
(4) As El Nino results in higher oceanic temperatures, the number of fish comes down due to lower plankton production.
2. Which of the following data will be helpful in the prediction of El Nino?
(1) Daily temperatures recorded in coastal areas.
(2) Daily atmospheric pressure levels for one previous year.
(3) A data bank of sea-surface temperatures.
(4) All of the above
3. The frequency of occurrence of El Nino, approximately, is
(1) once every year
(2) about three times in a century
(3) once in three years

	(4) once every seven years 4. The passage talks about (1) the factors that can trigger phenomena like El Nino. (2) the advances being made in the field of meteorological predictions through a greater study of ocean-atmosphere interactions. (3) phenomenon like El Nino that affect the planet's climate. (4) the predisposing factors that result in El Nino and the ways to construct them.				
5	Assuming yourself participating in a debate write four arguments in favour or against on one of the following topics- i-Robots increase people's quality of life. ii-Modern parenting has made kids impolite and less developed.	4	3	E	C

Section – B

Attempt All Questions 3 X 5 = 15

Marks

No.	Detail of Question	Marks	CO	BL	KL
6	i-What tips will you give to your friend about proper use of eye contact in Presentation? Choose the correct option- 1. Deciding it was time to sit down and begin a novel, the would-be writer spent the first several hours deciding upon a suitable ..... A.extempore B.nom de plume C.ad hoc D.vis à vis 2. Central High School is my ..... A. au revoir B. vide C. sangfroid D. alma mater 3. He has a ..... of knowledge when it comes to human psychology.	1+2	5,4	U,R	F

	A. plethora B. ad hoc C. pro bono D. phobia 4. The restaurant had a delightful ..... A. gratis B. status quo C. ambiance D. gourmet				
7	i. Having encountered several challenges, they _____ unchartered territories now. a. hated to explore b. explore c. are exploring d. hate Choose the most suitable option: A. only a B. only d C. Both c and d D. Both b and d ii. The project _____ undergone significant modifications. Use the most suitable primary auxiliary verb: A. is B. is doing C. has D. was iii. He _____ have completed the task by now. Choose the least suitable modal auxiliary verb: A. ought to B. must C. should D. can iv. _____ a marathon requires months of dedicated training. Use a non-finite verb as a noun: A. Winning	3	I	A	F

	B. Run for C. Participating D. Training v. The decision _____ these changes was not taken lightly. Choose the most appropriate option: A. was taken to implement B. is to implement C. to implement D. implement vi. I need to go to the doctor as I think that I might have a chronic disease. Which of the following verbs is NOT used in the sentence given below? A. primary auxiliary verb B. main verb C. semi modal auxiliary verb D. non-finite verb				
8	**Read the following passage. Identify sentence type-** i-These interfaces accept requests written in English or some other language and attempt to understand them. ii. A natural language interface usually has its own schema, which is similar to the database conceptual schema, as well as a dictionary of important words. iii. The natural language interface refers to the words in its schema, as well as to the set of standard words in its dictionary, to interpret the request. iv. If the interpretation is successful, the interface generates a high-level query corresponding to the natural language request and submits it to the DBMS for processing; otherwise, a dialogue is started with the user to clarify the request. v. The capabilities of natural language interfaces have not advanced rapidly. vi. Today, we see search engines that accept strings of natural language (like English or Spanish) words and match them with documents at specific	3	1	U	F

	sites (for local search engines) or Web pages on the Web at large (for engines like Google or Ask). i A. Simple B. Complex C. Negative D. Compound ii. A. Simple, complex B. Complex, compound C. Simple D. Compound, complex iii. A. Simple, complex B. Complex, compound C. Simple D. Compound iv A. Simple, complex B. Complex, compound C. Simple D. Compound, complex v. A. Simple B. Complex C. Complex, Compound D. Compound vi A. Simple, complex B. Complex, compound C. Simple D. Compound, complex				
9	Read the following passage and answer the questions that follow- In the forsaken outskirts of the bustling city, where dim lights flicker and children fear to tread at night, a mysterious figure appeared. It was the detective known for solving	3	1	A	F

	perplexing cases. The police department was instructed to carefully examine the evidence and submit a preliminary investigation report. The detective could help piece together fragments of a puzzle that seemed unsolvable. The witnesses whispered about the rumours surrounding the young man's demise as the detective hurriedly typed something on her phone. Having hurt her foot in the day, the detective struggled to jump over the stretched wires. The pine trees stood alongside the road like students lined up in a parade but they appeared formidable, not fragile. A suffocating smell filled the air as the night wore on and the city held its breath awaiting the revelation of the truth. In the above paragraph, identify the following: i. Two past participles used as adjectives ii. a. One present participle used as a verb, b. One perfect participle iii. a. One linking verb, b. One modal auxiliary verb iv. Two transitive verbs along with their objects v. Two intransitive verbs vi. one bare infinitive and one split infinitive				
10	Frame sentences as directed- i- Inquire about the objective of your project. ii- Show surprise on receiving an award. iii-Share any experience in a complex sentence. iv-Instruct your friend the way to the hostel in simple sentence. v-Congratulate your friend on his placement. vi-Write a declarative sentence about GLA University.	3	2	C	F

Section – C

Attempt All Questions 5 X 3 = 15

Marks

No.	Detail of Question	Marks	CO	BL	KL
11	The sentences (labeled 1, 2, 3, 4, 5) given in this question, when properly sequenced, form a coherent paragraph. Each sentence is labeled with a number. Decide on the proper order for the sentences and key in this sequence of five numbers as your answer. i- 1. The implications of retelling of Indian stories, hence, takes on new meaning in a modern India. 2. The stories we tell reflect the world around us. 3. We cannot help but retell the stories that we value — after all, they are never quite right for us — in our time. 4. And even if we manage to get them quite right, they are only right for us — other people living around us will have different reasons for telling similar stories. 5. As soon as we capture a story, the world we were trying to capture has changed. A. 25341 B. 32145 C. 12345 D. 42315 ii- 1. People with dyslexia have difficulty with print-reading, and people with autism spectrum disorder have difficulty with mind-reading. 2. An example of a lost cognitive instinct is mind-reading: our capacity to think of ourselves and others as having beliefs, desires, thoughts and feelings. 3. Mind-reading looks increasingly like literacy, a skill we know for sure is not in our genes, since scripts have been around for only 5,000-6,000 years. 4. Print-reading, like mind-reading varies across cultures, depends heavily on certain parts of the brain, and is subject to developmental disorders. A. 3214 B. 2341 C. 4321 D. 1234 iii- 1. If you've seen a little line of text on websites that says something like "customers who bought this also enjoyed that" you have experienced this collaborative filtering firsthand.	5	2	U	C

2. The problem with these algorithms is that they don't take into account a host of nuances and circumstances that might interfere with their accuracy.
3. If you just bought a gardening book for your cousin, you might get a flurry of links to books about gardening, recommended just for you! – the algorithm has no way of knowing you hate gardening and only bought the book as a gift.
4. Collaborative filtering is a mathematical algorithm by which correlations and cooccurrences of behaviors are tracked and then used to make recommendations.

A. 3214 B. 4123 C. 1234 D. 2314

iv-

1. It advocated a conservative approach to antitrust enforcement that espouses faith in efficient markets and voiced suspicion regarding the merits of judicial intervention to correct anticompetitive practices.
2. Many industries have consistently gained market share, the lion's share – without any official concern; the most successful technology companies have grown into veritable titans, on the premise that they advance 'public interest'.
3. That the new anti competitive risks posed by tech giants like Google, Facebook, and Amazon, necessitate new legal solutions could be attributed to the dearth of enforcement actions against monopolies and the few cases challenging mergers in the USA.
4. The criterion of 'consumer welfare standard' and the principle that antitrust law should serve consumer interests and that it should protect competition rather than individual competitors was an antitrust law introduced by, and named after, the 'Chicago school'.

A.1234 B. 4123 C. 3124 D. 4321

v-

1. The woodland's canopy receives most of the sunlight that falls on the trees.
2. Swifts do not confine themselves to woodlands, but hunt wherever there are insects in the air.
3. With their streamlined bodies, swifts are agile flyers, ideally adapted to twisting and turning through the air as they chase flying insects – the creatures that form their staple diet.

	4. Hundreds of thousands of insects fly in the sunshine up above the canopy, some falling prey to swifts and swallows. A.1234 B. 1324 C. 4321 D. 1432				
12	Rewrite the following sentences in minimum possible words so as to make them as brief as possible- i. As of 4 August, students will be selected using processes which have been created in accordance with legal requirements. These processes will be used along with our diversity strategy and all of its relevant policies about student selection. ii. The Passport Office is replacing free of charge until the dates specified any current passport lost, damaged or destroyed as a result of the recent floods. Eligible applicants should contact the Passport Information Service (PIS) on 998 9988 for further information. iii.If an agency or organization breaches the privacy principles, our office may investigate the matter. Individuals can also make a privacy complaint to us about an agency or organisation if they think their information has been mishandled. iv. Aid Agency works to bring immediate assistance to people whose lives have been devastated by natural disasters and conflict. Through our work we provide emergency relief in the initial stages of the disaster, and we also provide continued support to affected communities as they rebuild and rehabilitate towards a sustainable future. v. It is apparent that the tragedy could have been avoided if the company had talked to workers regarding hazards which existed in the workplace.				
13	You have to give a presentation analyzing the data on the topic 'Is Hybrid Work Sustainable?' Prepare a questionnaire for the same. (Minimum 5 questions & personal information detail)	5	5	An	C

Course Name: Data Visualization

Course Outcome

CO1- Have a solid understanding of the data analysis process and the significance of EDA in data science.

CO2- Be proficient in using Python to perform data visualization and exploratory analysis.

CO3- Gain hands-on experience with popular data visualization libraries like Matplotlib, Seaborn, and Plotly.

CO4- Be equipped with the skills to create an interactive dashboard using plotly and geospatial data analysis.

CO5- Be able to communicate complex data insights effectively through compelling data Visualizations

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End Term Examination, Odd Semester 2023-24
BTech (Hons) CSE, Year: II, Semester: III
BCSS 0051 & Data Visualization

Time: 3 Hours **Maximum Marks: 45**

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No	Detail of Question	Marks	CO	BL	KL
1	i. What is the calculated field in Tableau? (1M) ii. Why filters are required in Tableau. (1M) iii. How you will create a calculated field show with an example. (2M)	4	CO1	R	F
2	i. Why is the time series algorithm required? ii. What is the difference between ARIMA and SARIMA. iii. How to convert non stationary data to stationary data. iv. How to find trend and seasonal components in the time series data.	4	CO3	U	C
3	What is geospatial analysis of data? Write a python code for plotting choropleth by using plotly. Assume that the datafile is 'election' and the geojson file is 'election_geojson'. The election data contains columns like winner and district. The election_json file contains geometry and coordinates.	4	CO1	A	P
4	Explain data analysis process steps in detail.	4	CO1	R	C
5	A marketing team wants to compare the performance of two different advertising strategies over the past year. Suggest a suitable data visualization method to highlight key insights.	4	CO5	C	M

Section – B

*Attempt **All** Questions* 3 X 5 = 15 Marks

No	Detail of Question	Marks	CO	BL	KL
6	A manager wants to understand the trend in website traffic over the past month. Describe the type of chart or graph you would use and justify your choice.	3	CO5	C	M
7	Define the term data outliers? Mention the graphs which are used to find the outliers and write the python code for the same.	3	CO1	U	C
8	Why is the 'wordcloud' visualization required? Write a python code to plot a wordcloud. Assume the dataframe is 'data' and column name 'review'.	3	CO3	A	C
9	Mention the differences between univariate, bivariate and multivariate analysis. Explain univariate, bivariate and multivariate graphs with a python code.	3	CO3	R	F
10	As a data analyst, you are tasked with presenting the correlation between employee engagement scores and project success rates. What type of visualization would effectively convey this relationship, and why?	3	CO2	C	M

Section – C

*Attempt **All** Questions* 5 X 2 = 10 Marks

No	Detail of Question	Marks	CO	BL	KL
11	i. What is an interactive dashboard? Write python code to plot a 3d graph by using plotly. Assume that the dataframe is 'admission' and it has features like, CGPA, Research, SOP, ChanceOfAdmit(output). (2M) ii. Mention the differences between matplotlib, seaborn and plotly. (2M) iii. Why is a waffle chart required?	5	CO4	An	P
12	Consider the following Dataframe "company" which has the following columns: Emp_Age= [Nan,18,20,35,40,250,27, Nan] Emp_Salary= [50000,30000,40000,45000, Nan,37000,45000,34000]	5	CO5	E	M

	JobCode=['IT', 'Sales', 'Testing', 'IT', 'Admin', 'Marketing', 'Sales', 'IT'] EmpRating=[4,1,7,8,9,3,6,8] Promotion= [No, No, yes, yes, yes, No, yes, yes] Note: output variable is "Promotion". By using the above data frame 'data' answer the following questions. Write a python code, i. Perform simple EDA and analyze how all the input parameters affect the outcome with appropriate graphs. ii. Use appropriate methods to replace Nan values. iii. Apply proper data engineering techniques to normalize the data. iv. Apply appropriate data engineering steps to convert categorical data to numerical data. v. Which feature is very important to get promotion for the employees Write your conclusion.				

Course Name: Probability and Applied Statistics

Course Outcome

CO1-List different kind of statistics for data analysis.
CO2-Differentiate measure of central tendency and measure of variability.
CO3-Define Normalization and Standardization.
CO4-Conceptualize Probability distribution
CO5-Apply statistics in various research methods.

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End Term Examination, Odd Semester 2023-24
B. TECH(H), Year II, Semester III
BCSS 0052: Probability and Applied Statistics

Time: 3 Hours

Maximum Marks: 45

Section – A

Attempt All Questions

4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	You have collected data on the monthly sales of a product for the past year. The sales data (in thousands of units) is as follows: 15, 18, 12, 14, 16, 20, 22, 25, 19, 17, 16, 13, 15, 18, 12, 14, 16, 20, 22, 25, 19, 17, 16, 13 1. Create a histogram for the given sales data with five bins. Clearly label the axes and provide insights into the distribution of sale. 2. Based on your visualizations, describe the central tendency of the sales data. 3. Discuss any outliers or notable features in the data that are evident from your visualizations. 4. If you were to communicate the sales performance to a non-technical audience, what key insights would you share based on the histograms, bar chart, and frequency plot?	4	CO1	A	C
2	Data on the monthly incomes (in thousands of Rs.) of a sample of individuals. The income data is as follows: 42, 38, 45, 50, 40, 42, 36, 44, 42, 3942, 38, 45, 50, 40, 42, 36, 44, 42, 39 1. Calculate the mean (average) monthly income for the given dataset. 2. Determine the median monthly income for the given dataset.	4	CO2	U	P

	3. Identify the mode (if any) in the monthly income distribution. 4. Discuss any implications or insights that the mean, median, and mode provide about the distribution of incomes.				
3	Among 157 Indian men, the mean systolic blood pressure was 146 mm Hg with a standard deviation of 27. We wish to know if on the basis of these data, we may conclude that the mean systolic blood pressure for a population of Indian is greater than 140. [Use $\alpha = 0.01$ and $Z_{0.99} = 2.33$] OR The heights of college students in a city are normally distributed with S.D. 6 cms. A sample of 100 students has mean height 158 cms. Test the hypothesis that the mean height of college students in the city is 160 cms. [$Z_{0.95} = 1.96$ at 5% level of significance]	4	CO5	An	F
4	Explain the key concepts and steps involved in hypothesis testing. Provide a detailed overview of the process, including the formulation of hypotheses, the choice of significance level, the selection of a test statistic, and the interpretation of results. Additionally, discuss the importance of $p-values$ and the role they play in hypothesis testing.	4	Co4	R	F
5	For each of the following, state whether you would expect to find a positive, a negative, or no correlation between the two variables. Then, suggest a probable reason for the relationship. 1. The number of hours spent studying and achievement on an exam. 2. The number of years remaining to retirement and the value of an individual's savings and RRSP (registered retirement savings plan). 3. A person's level of physical fitness and the chance they enjoy action/adventure movies	4	CO3	An	P

Section – B

*Attempt **All** Questions* 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL

6	A random sample of 10 boys had the following I.Q.'s 70, 120, 110, 101, 88, 83, 95, 98, 107, 100. Do these dates support the assumption of a population mean I.Q.'s of 100? Find a reasonable range in which most of the mean I.Q. values of samples of 10 boys lie. [5% level of significance tabulated value = 2.262]	3	5	An	C
7	The theory predicts the proportion of beans, in the four groups A, B, C, and D should be 9: 3 : 3 : 1. In an experiment with 1600 beans the numbers in the four groups were 882, 313, 287, and 118. Does the experimental result support the theory. [5% LOS = 7.81]	3	5	An	C
8	The following data gives the age and blood pressure (BP) of 10 sports persons. Name : A B C D E F G H I J Age (X) : 42 36 55 58 35 65 60 50 48 51 BP (Y) : 98 93 110 85 105 108 82 102 118 99 1. Find regression equation of Y on X and X on Y (Use the method of deviation from arithmetic mean) 2. Find the correlation coefficient (r) using the regression coefficients. 3. Estimate the blood pressure of a sports person whose age is 45.	3	5	E	P
9	A computer while calculating the correlation coefficient between the variable X and Y obtained the following results: $N = 30$; $\Sigma X = 120$ $\Sigma X^2 = 600$ $\Sigma Y = 90$ $\Sigma Y^2 = 250$ $\Sigma XY = 335$ It was, however, later discovered at the time of checking that it had copied down two pairs of observations as: (X, Y) : (8, 10) (12, 7) While the correct values were: (X, Y) : (8, 12) (10, 8). Obtain the correct value of the correlation coefficient between X and Y.	3	3	An	C
10	You work for a company that claims its new marketing strategy has increased the average monthly sales from \$15,000 to \$18,000. To test this claim, you collect data on monthly sales for a random sample of 25 months before the new strategy	3	2	A	P

	implementation and 30 months after. The sample means and standard deviations are as follows: • Before the New Strategy: ◦ Sample Mean: $14,500 ◦ Sample Standard Deviation: $2,000 ◦ Sample Size: 25 • After the New Strategy: ◦ Sample Mean: $18,200 ◦ Sample Standard Deviation: $2,500 ◦ Sample Size: 30 Perform a hypothesis test to determine if there is sufficient evidence to support the company's claim of an increase in average monthly sales. Use a significance level of 0.05. $Z_{0.95}$= 1.96 at 5% level of significance				

Section – C

*Attempt **All** Questions*

5 X 2 = 10 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	Analysing the test scores of students from two different schools, School A and School B. The test scores (out of 100) for a sample of students from each school are as follows: **School A**: 75, 80, 85, 88, 90, 92, 94, 98, 100, 75, 80, 85, 88, 90, 92, 94, 98, 100 **School B**: 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105 **1.** Construct a box-and-whisker plot for the test scores of both School A and School B on the same graph. **2.** Analyse and interpret the box-and-whisker plot. Discuss the central tendency, spread, and any potential outliers for each school.	5	2	A	F
12	Based on information given below about the treatment of 200 patients suffering from a disease, state whether the new treatment is comparatively superior to the conventional treatment.	5	5	E	C

	Favourable	Not Favourable	Total
New	60	30	90
Conventional	40	70	110
	100	100	200

Course Name: ADVANCED DATABASE MANAGEMENT SYSTEMS

Course Outcome

CO1: Understand the concept of database management systems and Relational database.
CO2: Identify the various data models used in database design.
CO3: Design conceptual models of a database using ER modeling for real-life applications and construct queries in Relational Algebra.
CO4: Create and populate a RDBMS for a real-life application, with constraints and keys, using SQL.
CO5: Select the information from a database by formulating complex queries in SQL.
CO6: Analyze the existing design of a database schema and apply concepts of normalization to design an optimal database.
CO7: Discuss the recovery system and be familiar with the introduction to the web database, distributed databases.
CO8: Design and implement the database system with the fundamental concepts of DBMS using Python.

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End Term Examination, Odd Semester 2023-24
B. Tech. [CS (Hons.)], Year: II, Semester: III
Subject Code: BCSC 0054
Subject Name: Advanced Database Management Systems

Time: 3 Hours **Maximum Marks: 45**

Instruction for students:
1. Mention clearly if any assumptions are being made.
2. No 'B' sheet will be given.

Section – A

*Attempt **All** Questions* 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Assume a case of an Automobile company that sells Motorcycles, Cars, Vans & Buses. Draw a Generalization – Specialization hierarchy diagram for a motor – vehicle sale. Also convert your diagram into tables.	4	3	C	C
2	Given the Relation R(ABCDEFGH) and Functional Dependencies as below: FD: {ABC→DE, E→GH, H→G, G→H, ABCD→EF} Identify the highest normal form of this relation.	4	4	R	P
3	Define Lock. Also explain 2PL protocol in details with the help of suitable diagram.	4	7	R	F
4	Answer the following: a. Define properties of the transaction b. State diagram of the transaction.	4	7	U	C

5	Consider the following table named, ***Subject_Info*** **Semester** | **Subject** | **Instructor** Semester 1 | Mathematics | Nivedita Semester 1 | Mathematics | Jatin Semester 1 | Programming | Jatin Semester 1 | Chemistry | Shivani Semester 2 | Programming | Kartik Decompose this relation into required number of sub relations so that the decomposed sub relations ensure lossless join decomposition.	4	6	An	C

Section – B

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Write a short note on following: a. Multivalued Dependency, & b. Fourth Normal Form	3	6	R	F
7	Differentiate between Homogeneous Distributed Databases and Heterogeneous Distributed Databases	3	7	R	F
8	Define following problems with respect to the concurrent execution of the transaction: a. Lost Update Problems, b. Dirty Read Problems	3	6	U	C
9	Is the following schedule S serializable? If yes, is it conflict serializable or view serializable? Identify. S: $R_1(C)$ $W_1(B)$ $R_2(B)$ $W_2(A)$ $W_1(A)$ $W_2(A)$	3	7	R	C
10	Consider the concurrent execution of two transactions T1 & T2. If the initial values of x, y, M and N are 200, 100, 10 and 20 respectively; Compute the final values of x and y. **T1** | **T2** R (x) | x = x – N | | R (x) | x = x + M W (x) | R (y) | | W (x) y = y + N | W (y) |	3	7	E	P

Section – C

*Attempt **All** Questions* 5 X 2 = 10 Marks

<table>
<tr><th>No.</th><th>Detail of Question</th><th>Marks</th><th>CO</th><th>BL</th><th>KL</th></tr>
<tr><td>11</td><td>Given, a database that has the relation schema as, CR (StudentName, CourseName).
An instance of the schema CR is given as below:
<table><tr><th>StudentName</th><th>CourseName</th></tr><tr><td>SA</td><td>CA</td></tr><tr><td>SA</td><td>CB</td></tr><tr><td>SA</td><td>CC</td></tr><tr><td>SB</td><td>CB</td></tr><tr><td>SB</td><td>CC</td></tr><tr><td>SC</td><td>CA</td></tr><tr><td>SC</td><td>CB</td></tr><tr><td>SC</td><td>CC</td></tr><tr><td>SD</td><td>CA</td></tr><tr><td>SD</td><td>CB</td></tr><tr><td>SD</td><td>CC</td></tr><tr><td>SD</td><td>CD</td></tr><tr><td>SE</td><td>CD</td></tr><tr><td>SE</td><td>CA</td></tr><tr><td>SE</td><td>CB</td></tr><tr><td>SF</td><td>CA</td></tr><tr><td>SF</td><td>CB</td></tr><tr><td>SF</td><td>CC</td></tr></table>The following queries are made on the database:
T1 ← $\pi_{\text{CourseName}}$ ($\sigma_{\text{StudentName = 'SA'}}$ (CR))
T2 ← CR ÷ T1
After performing these queries, Show the output of T1 and T2 respectively.</td><td>5</td><td>3</td><td>E</td><td>P</td></tr>
<tr><td>12</td><td>Consider the following two Transactions:
T1: Read (A);
Read (B);
if A = 0 then B: = B + 1;
Write (B);
T2: Read (B);
Read (A);
if B = 0 then A: = A + 1;
Write (A);
Add the appropriate lock and unlock instructions to these transactions, so that they observe the 2PL protocol.
Can the execution of these transactions (T1 & T2) result in a deadlock? Justify your answer.</td><td>5</td><td>7</td><td>E</td><td>P</td></tr>
</table>

Course Name: English For Professional Purposes I

Course Outcome

CO1- create structurally effective, grammatically accurate and contextually appropriate written and spoken pieces for communication both independently and collaboratively.

CO2-display competence in all aspects of language learning not limited to listening, speaking, reading and writing.

CO3-develop independent perspectives and points of view and express them in a cohesive and meaningful way.

CO4-produce a repertoire of register-specific and academic vocabulary appropriate in both oral and written communication.

CO5-use language as an effective tool of communication for various contexts in both academic and professional setting.

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End Term Examination, Odd Semester 2023-24

B.Tech II Year, III Semester (All Branches) & MCA I Year, I Semester

BELH 1003 English for Professional Purposes I

Time: 3 Hours **Maximum Marks: 50**

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Mark	CO	BL	KL
1	i-Identify the grammatical function of the word fast in each of the following sentences 1- By the time I got home, the children were already fast asleep. A. Noun B. Verb C. Adverb D. Preposition 2- Some prisoners began a fast to protest against the appalling conditions A. Verb	4	1	A	F

B. Adjective
C. Noun
D. Interjection

i-Kelly has great confidence _____ her children's ability.
Choose the correct preposition.
A. on
B. at
C. for
D. in

iii. Unfortunately, many online courses do not provide students _____ the basic skills they need.
A. with
B. for
C. on
D. in

iv. They hope to can arrive before the start of the conference.
Identify the unnecessary word in the sentence and its part of speech.
A. can- linking verb
B. to- infinitive
C. the (second) – Definite article
D. can - Auxiliary Verb

v. J: I don't have any money.
K: I don't have no money.
L: I don't have money.
The grammatically correct sentences are:
A. J and K
B. K and L
C. J and L
D. J, K and L

vi. Leap year falls in _____ forth year.
A. Each
B. Every
C. One

	D. Any vii. He showed _____ concern for his nephew. A. the little B. a few C. little D. a little viii. Whether I shall turn out to be the hero of my own life, or whether that station will be held by anyone else, _____ pages must show. A. these B. this C. none D. third ix. _____truth must be spoken. A. a B. the C. no article D. Both A and B. x.In which year were both these articles circulated to all the students? Identify the determiners. A. to, all, these, both. B. both, the, all, in. C. to, year, all, the. D. these, the, both, all xi. Linear communication is _____ one-way process conducted in ____ straight line towards ____ main point. Fill in the blanks with correct determiners. A. a, the, the B. the, a, a C. a, the, a D. a, a, the xii. Most men and women will grow up to love _____				

servitude and will never dream of revolution.

Fill in the blank with the correct determiner.

A. our
B. there
C. their
D. Many

xiii-The manager said to me, "Complete it by tomorrow."

A. The manager requested me to complete it by tomorrow.
B. The manager instructed me to complete it by the next day.
C. I was ordered by the manager to complete the work by the next day.
D. The manager ordered me to complete the work by the next day.

xiv- J) Today it was first time we play such games.
k) Bacteria can kill you into a few days.
L) I am leaving a call at 10:00 AM.

The grammatically correct sentence is:

A. J and K
B. K and L
C. J and L
D. None of the above

xv- J) I will buy some mangoes.
K) I will not buy some mangoes.
L) A little knowledge is a dangerous thing.
M) A darkest cloud has a silver lining.

Which of the following sentences have the correct use of determiners?

A. J and L only
B. K, L and M are correct
C. Only J is correct
D. Only M is correct

xvi- He gave a talk _____ the causes of Dengue.

A. on

	B. for C. at D. of				
2	i He is an *absolutely honest* politician. A. Noun Phrase B. Prepositional Phrase C. Adjective Phrase D. Adverb Phrase ii.*In the realm of quantum mechanics*, researchers delve into the mysteries of subatomic particles. A. Prepositional Phrase B. Verb Phrase C. Adverb Phrase D. Independent Clause iii. The arrival *of the train* has been delayed. A. Adverb Phrase B. Prepositional Phrase C. Noun Phrase D. Verb Phrase iv. *All my sons* are good football players. A. Noun Phrase B. Adjective Phrase C. Adverb Phrase D. Verb Phrase v. It *has been raining* continuously for the last seven days. A. Verb Phrase B. Adverb Phrase C. Adjective Phrase D. Noun Phrase vi. The men in the red scarves dance *quite merrily* in the sun. A. Verb Phrase B. Noun Phrase C. Adverb Phrase D. Adjective Phrase vii.The people look solemn, yet they go to work every day. A. Independent Clause followed by a Dependent Clause. B. Dependent Clause followed by an Independent Clause. C. 2 Independent Clauses	4	1	U	F

D. 2 Dependent Clauses

viii. *Although the train arrived on time*, we still missed it.

A. Dependent Clause
B. Independent Clause
C. Prepositional Phrase
D. Verb Phrase

ix. Jane's reputation as a/an _____ grew as she consistently prioritized her own needs over those of her colleagues.

A. egoist
B. versatile
C. flamboyant
D. selfish

x. Before responding, he took a moment to _____, the impact of his words on the conversation.

A. contemplate
B. introspect
C. adapt
D. accustom

xi. When choosing mutual funds, individuals often prefer those with a ____ history of reliable returns and performance.

A. bankrupt
B. bankable
C. prominent
D. amateur

xii.The old computer system became _____ after the company upgraded to the latest technology.

A. famous
B. redundant
C. obsolete
D. obliterate

xiii. The Indian Army _____ INS Viraat in the year 2017.

A. decommissioned
B. simulate
C. reinstitute
D. rehabilitate

xiv. The math _____ effortlessly solved complex problems that left even seasoned mathematicians impressed.

A. novice
B. servile

	C. gifted D. prodigy xv. The jury decided on the capital punishment _____ A. impromptu B. solely C. unanimously D. servile xvi. The _____ of the situation was that the more she tried to control time, the more it seemed to slip away from her grasp. A. paradox B. agenda C. veteran D. fanciful				
3	i. Rewrite the following *para* to make it coherent. "Limited investment in the housing sector makes it practically impossible to allocate sufficient resources for urban dwellers' housing needs. A high rate of urban population growth has increased the country's needs for housing. A small group of city officials has laid out a new plan to combat the crisis. A solution to the housing-shortage problem is a vital policy issue here. The housing problem has grown in the last twenty years." ii. Fill in the blanks with the most suitable transitions from the list. Use each transition only once: in spite of, thanks to, although, similar to, like, also, in addition to, whereas, compared to, A._____ we have scientific evidence that women should not smoke when they are pregnant, many future mothers ignore this danger. These mothers endanger the physical health of their babies. B-_____, they could be contributing to even more problems in the future. C- _____ the known problems, recent studies have linked mothers' smoking during their pregnancy with sons who go on to become criminals. Researchers in Finland followed over 5000 males and their mothers from the time the mothers were six months pregnant until the young men	2+2	2	A	C

	were 28 years old. D- _____ sons whose mothers did not smoke during pregnancy, those young men whose birth mothers did smoke were more than twice as likely to commit criminal acts. The researchers theorized that smoking during pregnancy may alter the chemical balance in the baby's brain.				
4	Read the following passages and answer the questions that follow: About once in every seven years, the ocean surface off the coast of Peru warms up. This cuts the normal enriching of nutrient-rich cold water. Plankton production is drastically reduced. This phenomenon, known as El Nino (the child) because it starts during the Christmas season, usually lasts for up to a year. Occasionally, it goes on for long. Three times this century, it has persisted into a third year. The latest of these prolonged episodes have been blamed for much more than its effect on the Peruvian fishery. It certainly seems to have played a part in droughts in normally humid Indonesia as well as those that brought catastrophic fires to the outskirts of Sydney. It has been blamed for storms and landslides in coastal regions of Peru and Equator and is associated with drought in north-eastern Brazil. Its influence may stretch as far as western Europe where the recent winter brought heavy rainfall and flooding. El Nino is not the only ocean phenomenon to affect the weather. To understand how this might be done, it is useful to think about how the weather is forecast and about what makes it predictable. Weather forecasting uses two types of techniques, both of which rely on observations of what is going on at and shortly before the time of the forecast. To supply this information, national authorities operate extensive land, sea and air based observation networks. These are coordinated under the United Nations World Meteorological Organisation. There is an effective and almost instantaneous worldwide exchange of the information gathered. The first, classical forecasting method compares the set of observations with experience and bases the forecast on what has happened in the past from a similar starting point. The second, only possible since the advent of fast computers, applies the laws of physics through a numerical model of the atmosphere to	4	2	U	F

predict how the weather will develop from the observed starting point. This method is the basis of modern weather forecasting. But neither method will produce useful predictions for more than a few days ahead. This is because the behaviour of the atmosphere is governed in the medium term by energy exchanges with the land and ocean. The latter is much more important as the sea surface stores energy which it exchanges with the atmosphere in a surface layer 100 m or so thick. The energy storage capacity of the layer is many times greater than that of the whole atmosphere. This had prompted the suggestion that longer term forecasting might be practicable if a coupled computer model of the atmosphere and ocean could be constructed.

A great deal of research has been put into this with some success but progress has been limited by two factors. The first is that coupled ocean-atmosphere models require enormously greater computing capacity than is provided even by the super computers used in weather forecasting. The second is that not enough is known about the state of the ocean at any given time - there is no global observation network as there is for the atmosphere–or about the processes that govern the interactions. When these limitations have been overcome, it seems likely that coupled models will permit prediction of such climatological factors as frequency and intensity of rainfall for seasons and perhaps years ahead.

In the meantime, climatologists have begun to be able to predict the onset and consequences of phenomena such as El Nino using techniques like those used in the classical weather forecasting method. These examine the condition of the ocean at a given instant and, by comparison with past experience, attempt to predict in a statistical way how the ocean or the atmosphere—and hence the climate—are likely to behave for the next few months. As with the weather forecasts, this depends critically on the existence of a bank of past experience, in this case, particularly of sea-surface temperatures. A key contribution to this, recently completed by the UK Meteorological Office, has been the analysis of many millions of sea-surface temperatures. This has led to the publication of the first globally complete monthly fields of sea-surface temperature from 1871 to the present day. This

type of information may be used to predict events such as El Nino.

Various scientific groups have tried to do so with varying degrees of success. In the meantime, Meteorological Office scientists have also compared the temperature fields statistically with climatological factors. They have shown that, particularly in the tropics, there are significant correlations between sea-surface temperature anomalies and climate statistics. This does not necessarily mean that one causes the other, though some degree of direct linkage seems likely. But it does open up the possibility of predicting short-term climate fluctuations.

The UK Meteorological Office has approached this by a rigorous comparison between rainfall statistics in the Nordeste area of Brazil, whose crops can be seriously affected by drought, with contemporaneous sea-surface temperatures world-wide. This has revealed significant links, verified over the period 1901-85, between rainfall and sea-surface temperatures in the north and south tropical Atlantic and the western tropical Pacific, the area most strongly affected by El Nino.

1. Which of the following is NOT TRUE about 'El Nino'?
(1) It is the most important ocean phenomenon to affect a region's weather pattern.
(2) There seems to be a statistical link between sea-surface temperatures and the occurrence of El Nino.
(3) The consequences of El Nino can vary with the geographical position of the affected area.
(4) As El Nino results in higher oceanic temperatures, the number of fish comes down due to lower plankton production.

2. Which of the following data will be helpful in the prediction of El Nino?
(1) Daily temperatures recorded in coastal areas.
(2) Daily atmospheric pressure levels for one previous year.
(3) A data bank of sea-surface temperatures.
(4) All of the above

3. The frequency of occurrence of El Nino, approximately, is
(1) once every year
(2) about three times in a century
(3) once in three years

	(4) once every seven years 4. The passage talks about (1) the factors that can trigger phenomena like El Nino. (2) the advances being made in the field of meteorological predictions through a greater study of ocean-atmosphere interactions. (3) phenomenon like El Nino that affect the planet's climate. (4) the predisposing factors that result in El Nino and the ways to construct them.				
5	Assuming yourself participating in a debate write four arguments in favour or against on one of the following topics- i-Robots increase people's quality of life. ii-Modern parenting has made kids impolite and less developed.	4	3	E	C

Section – B

Attempt All Questions 3 X 5 = 15

Marks

No.	Detail of Question	Marks	CO	BL	KL
6	i-What tips will you give to your friend about proper use of eye contact in Presentation? Choose the correct option- 1. Deciding it was time to sit down and begin a novel, the would-be writer spent the first several hours deciding upon a suitable ..... A.extempore B.nom de plume C.ad hoc D.vis à vis 2. Central High School is my ..... A. au revoir B. vide C. sangfroid D. alma mater 3. He has a ..... of knowledge when it comes to human psychology.	1+2	5,4	U,R	F

	A. plethora B. ad hoc C. pro bono D. phobia 4. The restaurant had a delightful ..... A. gratis B. status quo C. ambiance D. gourmet				
7	i. Having encountered several challenges, they _____ unchartered territories now. a. hated to explore b. explore c. are exploring d. hate Choose the most suitable option: A. only a B. only d C. Both c and d D. Both b and d ii. The project _____ undergone significant modifications. Use the most suitable primary auxiliary verb: A. is B. is doing C. has D. was iii. He _____ have completed the task by now. Choose the least suitable modal auxiliary verb: A. ought to B. must C. should D. can iv. _____ a marathon requires months of dedicated training. Use a non-finite verb as a noun: A. Winning	3	1	A	F

	B. Run for C. Participating D. Training v. The decision _____ these changes was not taken lightly. Choose the most appropriate option: A. was taken to implement B. is to implement C. to implement D. implement vi. I need to go to the doctor as I think that I might have a chronic disease. Which of the following verbs is NOT used in the sentence given below? A. primary auxiliary verb B. main verb C. semi modal auxiliary verb D. non-finite verb				
8	**Read the following passage. Identify sentence type-** i-These interfaces accept requests written in English or some other language and attempt to understand them. ii. A natural language interface usually has its own schema, which is similar to the database conceptual schema, as well as a dictionary of important words. iii. The natural language interface refers to the words in its schema, as well as to the set of standard words in its dictionary, to interpret the request. iv. If the interpretation is successful, the interface generates a high-level query corresponding to the natural language request and submits it to the DBMS for processing; otherwise, a dialogue is started with the user to clarify the request. v. The capabilities of natural language interfaces have not advanced rapidly. vi. Today, we see search engines that accept strings of natural language (like English or Spanish) words and match them with documents at specific	3	1	U	F

	sites (for local search engines) or Web pages on the Web at large (for engines like Google or Ask). i A. Simple B. Complex C. Negative D. Compound ii. A. Simple, complex B. Complex, compound C. Simple D. Compound, complex iii. A. Simple, complex B. Complex, compound C. Simple D. Compound iv A. Simple, complex B. Complex, compound C. Simple D. Compound, complex v. A. Simple B. Complex C. Complex, Compound D. Compound vi A. Simple, complex B. Complex, compound C. Simple D. Compound, complex				
9	Read the following passage and answer the questions that follow- In the forsaken outskirts of the bustling city, where dim lights flicker and children fear to tread at night, a mysterious figure appeared. It was the detective known for solving	3	1	A	F

	perplexing cases. The police department was instructed to carefully examine the evidence and submit a preliminary investigation report. The detective could help piece together fragments of a puzzle that seemed unsolvable. The witnesses whispered about the rumours surrounding the young man's demise as the detective hurriedly typed something on her phone. Having hurt her foot in the day, the detective struggled to jump over the stretched wires. The pine trees stood alongside the road like students lined up in a parade but they appeared formidable, not fragile. A suffocating smell filled the air as the night wore on and the city held its breath awaiting the revelation of the truth. In the above paragraph, identify the following: i. Two past participles used as adjectives ii. a. One present participle used as a verb, b. One perfect participle iii. a. One linking verb, b. One modal auxiliary verb iv. Two transitive verbs along with their objects v. Two intransitive verbs vi. one bare infinitive and one split infinitive				
10	Frame sentences as directed- i- Inquire about the objective of your project. ii- Show surprise on receiving an award. iii-Share any experience in a complex sentence. iv-Instruct your friend the way to the hostel in simple sentence. v-Congratulate your friend on his placement. vi-Write a declarative sentence about GLA University.	3	2	C	F

Section – C

*Attempt **All** Questions* 5 X 3 = 15

Marks

No.	Detail of Question	Marks	CO	BL	KL
11	The sentences (labeled 1, 2, 3, 4, 5) given in this question, when properly sequenced, form a coherent paragraph. Each sentence is labeled with a number. Decide on the proper order for the sentences and key in this sequence of five numbers as your answer. i- 1. The implications of retelling of Indian stories, hence, takes on new meaning in a modern India. 2. The stories we tell reflect the world around us. 3. We cannot help but retell the stories that we value — after all, they are never quite right for us — in our time. 4. And even if we manage to get them quite right, they are only right for us — other people living around us will have different reasons for telling similar stories. 5. As soon as we capture a story, the world we were trying to capture has changed. A. 25341 B. 32145 C. 12345 D. 42315 ii- 1. People with dyslexia have difficulty with print-reading, and people with autism spectrum disorder have difficulty with mind-reading. 2. An example of a lost cognitive instinct is mind-reading: our capacity to think of ourselves and others as having beliefs, desires, thoughts and feelings. 3. Mind-reading looks increasingly like literacy, a skill we know for sure is not in our genes, since scripts have been around for only 5,000-6,000 years. 4. Print-reading, like mind-reading varies across cultures, depends heavily on certain parts of the brain, and is subject to developmental disorders. A. 3214 B. 2341 C. 4321 D. 1234 iii- 1. If you've seen a little line of text on websites that says something like "customers who bought this also enjoyed that" you have experienced this collaborative filtering firsthand.	5	2	U	C

2. The problem with these algorithms is that they don't take into account a host of nuances and circumstances that might interfere with their accuracy.
3. If you just bought a gardening book for your cousin, you might get a flurry of links to books about gardening, recommended just for you! – the algorithm has no way of knowing you hate gardening and only bought the book as a gift.
4. Collaborative filtering is a mathematical algorithm by which correlations and cooccurrences of behaviors are tracked and then used to make recommendations.

A. 3214 B. 4123 C. 1234 D. 2314

iv-

1. It advocated a conservative approach to antitrust enforcement that espouses faith in efficient markets and voiced suspicion regarding the merits of judicial intervention to correct anticompetitive practices.
2. Many industries have consistently gained market share, the lion's share – without any official concern; the most successful technology companies have grown into veritable titans, on the premise that they advance 'public interest'.
3. That the new anti competitive risks posed by tech giants like Google, Facebook, and Amazon, necessitate new legal solutions could be attributed to the dearth of enforcement actions against monopolies and the few cases challenging mergers in the USA.
4. The criterion of 'consumer welfare standard' and the principle that antitrust law should serve consumer interests and that it should protect competition rather than individual competitors was an antitrust law introduced by, and named after, the 'Chicago school'.

A.1234 B. 4123 C. 3124 D. 4321

v-

1. The woodland's canopy receives most of the sunlight that falls on the trees.
2. Swifts do not confine themselves to woodlands, but hunt wherever there are insects in the air.
3. With their streamlined bodies, swifts are agile flyers, ideally adapted to twisting and turning through the air as they chase flying insects – the creatures that form their staple diet.

	4. Hundreds of thousands of insects fly in the sunshine up above the canopy, some falling prey to swifts and swallows. A.1234 B. 1324 C. 4321 D. 1432				
12	Rewrite the following sentences in minimum possible words so as to make them as brief as possible- i. As of 4 August, students will be selected using processes which have been created in accordance with legal requirements. These processes will be used along with our diversity strategy and all of its relevant policies about student selection. ii. The Passport Office is replacing free of charge until the dates specified any current passport lost, damaged or destroyed as a result of the recent floods. Eligible applicants should contact the Passport Information Service (PIS) on 998 9988 for further information. iii.If an agency or organization breaches the privacy principles, our office may investigate the matter. Individuals can also make a privacy complaint to us about an agency or organisation if they think their information has been mishandled. iv. Aid Agency works to bring immediate assistance to people whose lives have been devastated by natural disasters and conflict. Through our work we provide emergency relief in the initial stages of the disaster, and we also provide continued support to affected communities as they rebuild and rehabilitate towards a sustainable future. v. It is apparent that the tragedy could have been avoided if the company had talked to workers regarding hazards which existed in the workplace.				
13	You have to give a presentation analyzing the data on the topic 'Is Hybrid Work Sustainable?' Prepare a questionnaire for the same. (Minimum 5 questions & personal information detail)	5	5	An	C